

Descriptive Statistics

2025-11-07

1. Data Overview

1.1 Description of the Data Gathering Process

The data considered in the project is from OpenAlex. we use a pair-based keyword search to identify articles which use or construct game-theoretical models. The keywords we are taken from two lists: 1. A list of JEL codes defining the subject of study, i.e. game theory. 2. A list of terms describing game setup which ensures that we only cover papers employing or constructing game-theortic models. For each pair of keywords, we identify articles based on the appearance of keyword pairs in the title or abstract of the paper (maybe we also want to consider full texts? This would increase the number of articles drastically.) Currently, we focus only on articles and reviews written in English (Should we include books?). We drop articles from the corpus without any reference information. As the number of articles is rather low prior to 1980, we choose this year as the starting point for our analysis.

1.2 Search Terms

Table 1: List of Query Terms

Term	Term
game theory	equilibrium
game-theoretic	equilibria
cooperative game	solution concept
cooperative games	solution concepts
noncooperative game	payoff matrix
noncooperative games	pay-off matrix
non-cooperative game	payoff matrices
non-cooperative games	pay-off matrices
stochastic game	strategy
stochastic games	strategies
dynamic game	strategic
dynamic games	preference
evolutionary game	preferences
evolutionary games	player
repeated game	players
repeated games	
bargaining theory	
matching theory	

1.3 Descriptive Summary of the Data

Table 2: Descriptive Summary

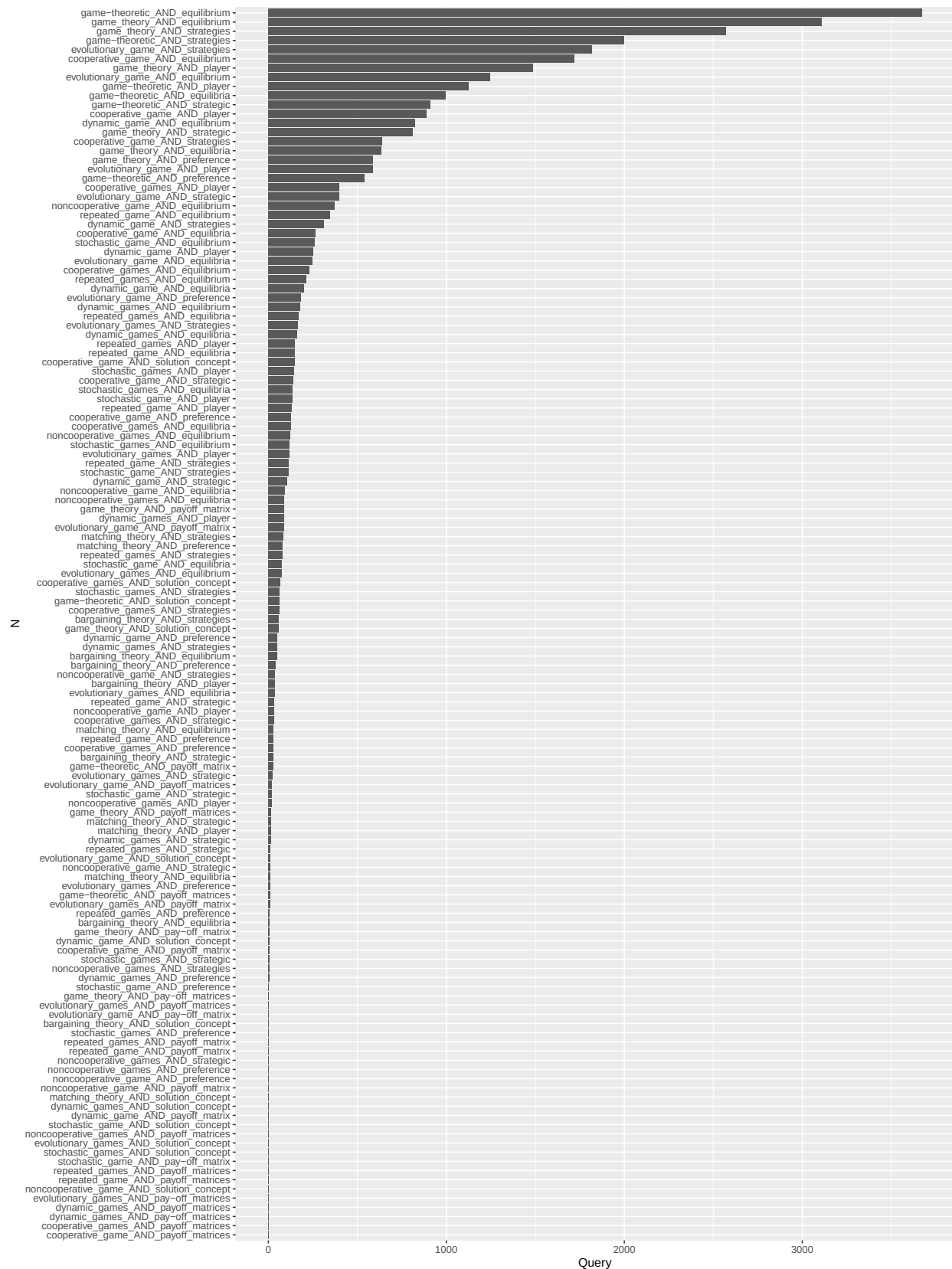
Description	
Articles	34676.00
Year (Begin)	1980.00
Year (End)	2024.00
Average Citations per Article	24.53
Authors	53383.00
Articles per Author	1.85
Authors per Article	2.85

1.4 Top Articles

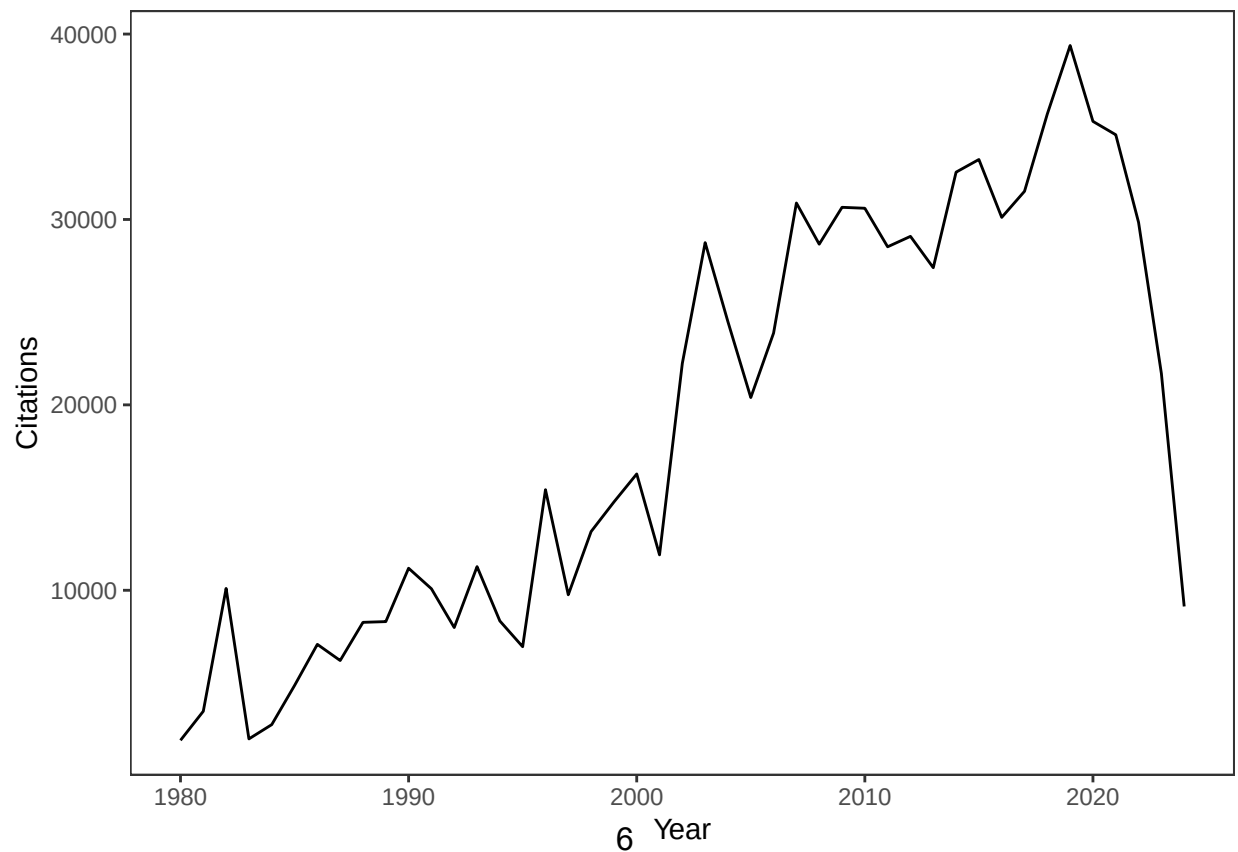
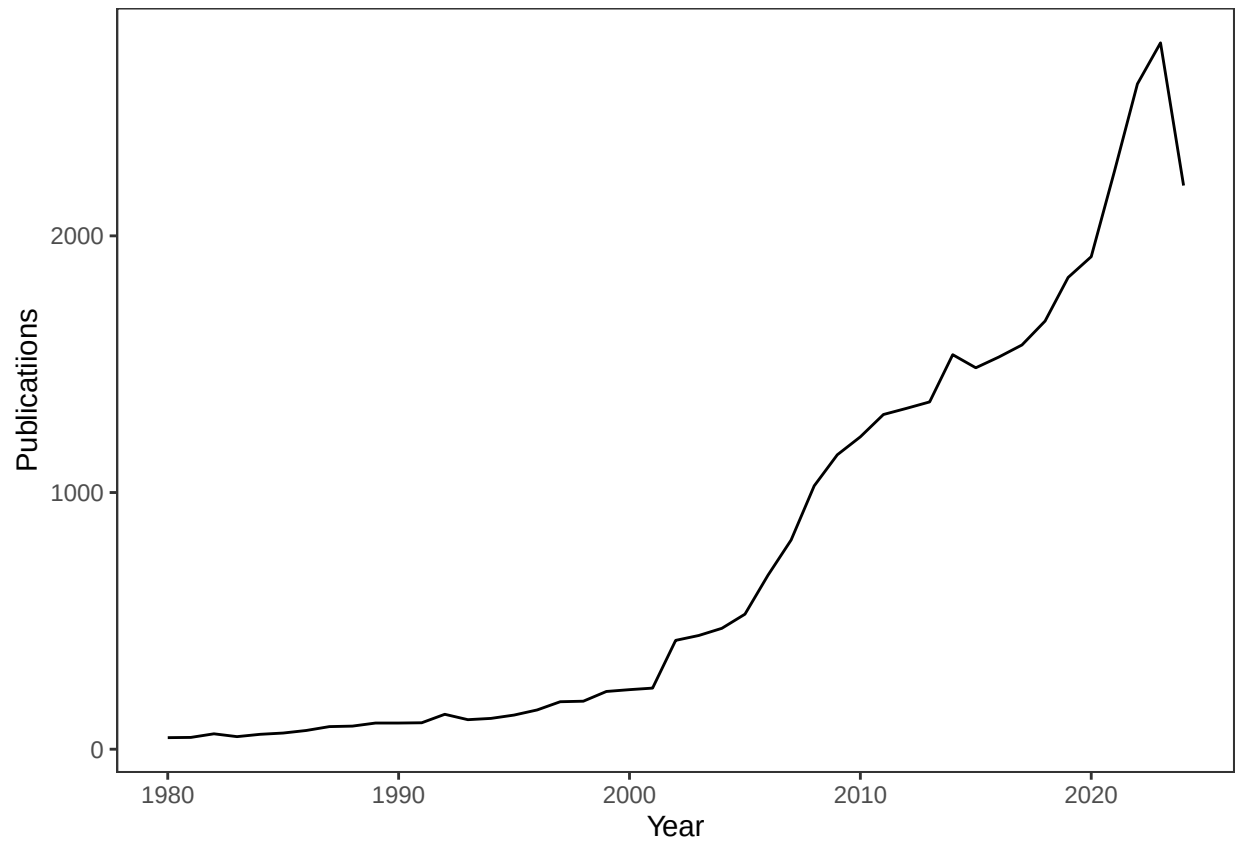
Table 3: Top Articles in Terms of Citations

Authors	Title	Year	Citations	Field
David Kempe;Jon Kleinberg;Éva Tardos	Maximizing the spread of influence through a social network	2003	7088	Physics and Astronomy
David M. Kreps;Robert Wilson	Reputation and imperfect information	1982	3101	Decision Sciences
György Szabó;Gábor Fáth	Evolutionary games on graphs	2007	2823	Social Sciences
Amir-Hamed Mohsenian-Rad;Vincent W. S. Wong;Juri Jatskevich;Robert Schober;Alberto Leon-Garcia	Autonomous Demand-Side Management Based on Game-Theoretic Energy Consumption Scheduling for the Future Smart Grid	2010	2711	Engineering
Arvind Parkhe	STRATEGIC ALLIANCE STRUCTURING: A GAME THEORETIC AND TRANSACTION COST EXAMINATION OF INTERFIRM COOPERATION.	1993	2581	Business, Management and Accounting
Xu Chen;Lei Jiao;Wenzhong Li;Xiaoming Fu	Efficient Multi-User Computation Offloading for Mobile-Edge Cloud Computing	2015	2552	Computer Science
David M. Kreps;Paul Milgrom;John Roberts;Robert Wilson	Rational cooperation in the finitely repeated prisoners' dilemma	1982	2512	Social Sciences
Drew Fudenberg;Eric Maskin	The Folk Theorem in Repeated Games with Discounting or with Incomplete Information	1986	2371	Decision Sciences
Hisashi Ohtsuki;Christoph Hauert;E. James Lieberman;Martin A. Nowak	A simple rule for the evolution of cooperation on graphs and social networks	2006	2155	Social Sciences
Charles F. Manski	Economic Analysis of Social Interactions	2000	2052	Economics, Econometrics and Finance
Martin Dufwenberg;Georg Kirchsteiger	A theory of sequential reciprocity	2003	1902	Social Sciences
Ido Erev;Alvin E. Roth	Predicting How People Play Games: Reinforcement Learning in Experimental Games with Unique, Mixed Strategy Equilibria	1998	1822	Social Sciences
NA	Rationalizability, Learning, and Equilibrium in Games with Strategic Complementarities	1990	1816	Decision Sciences
Daniel Friedman	Evolutionary Games in Economics	1991	1797	Economics, Econometrics and Finance
Barry R. Weingast	The Political Foundations of Democracy and the Rule of the Law	1997	1739	Economics, Econometrics and Finance

1.5 Number of Articles per Query

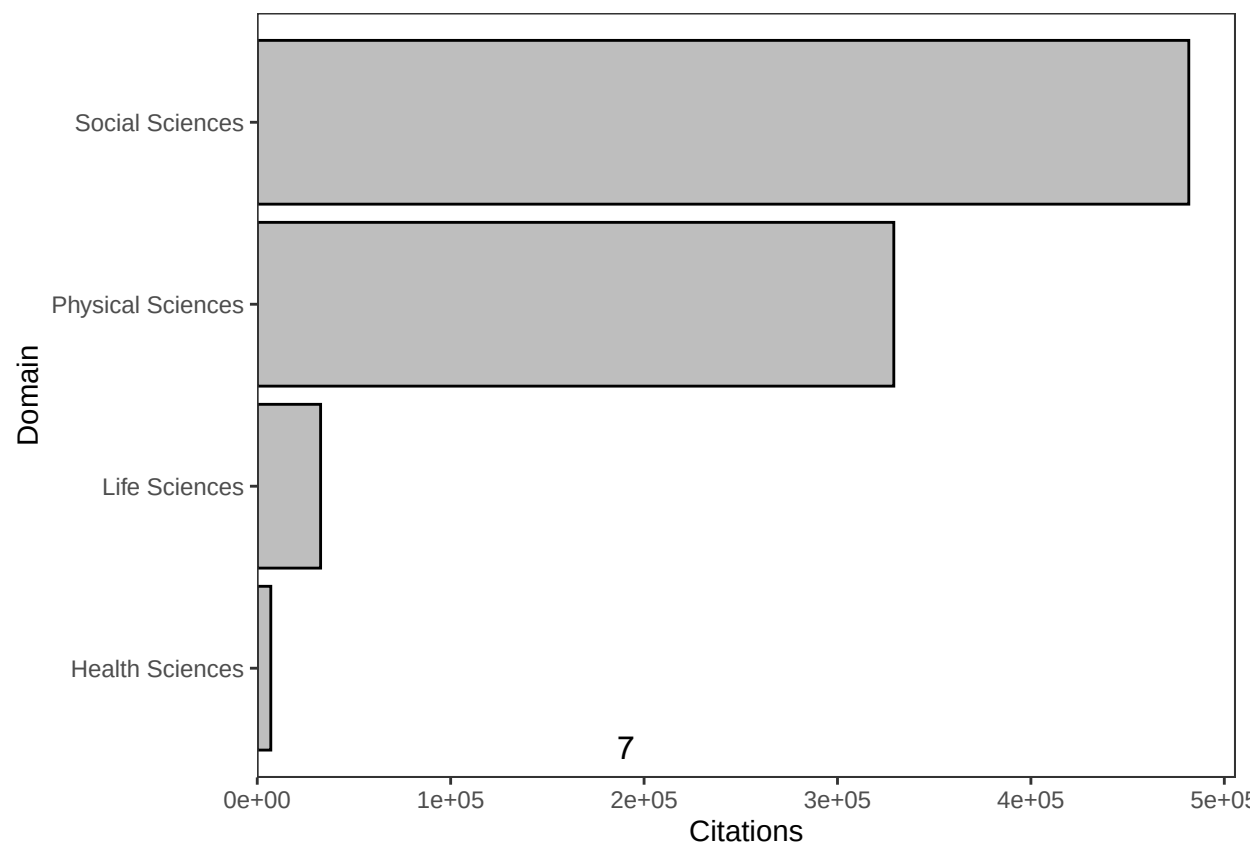
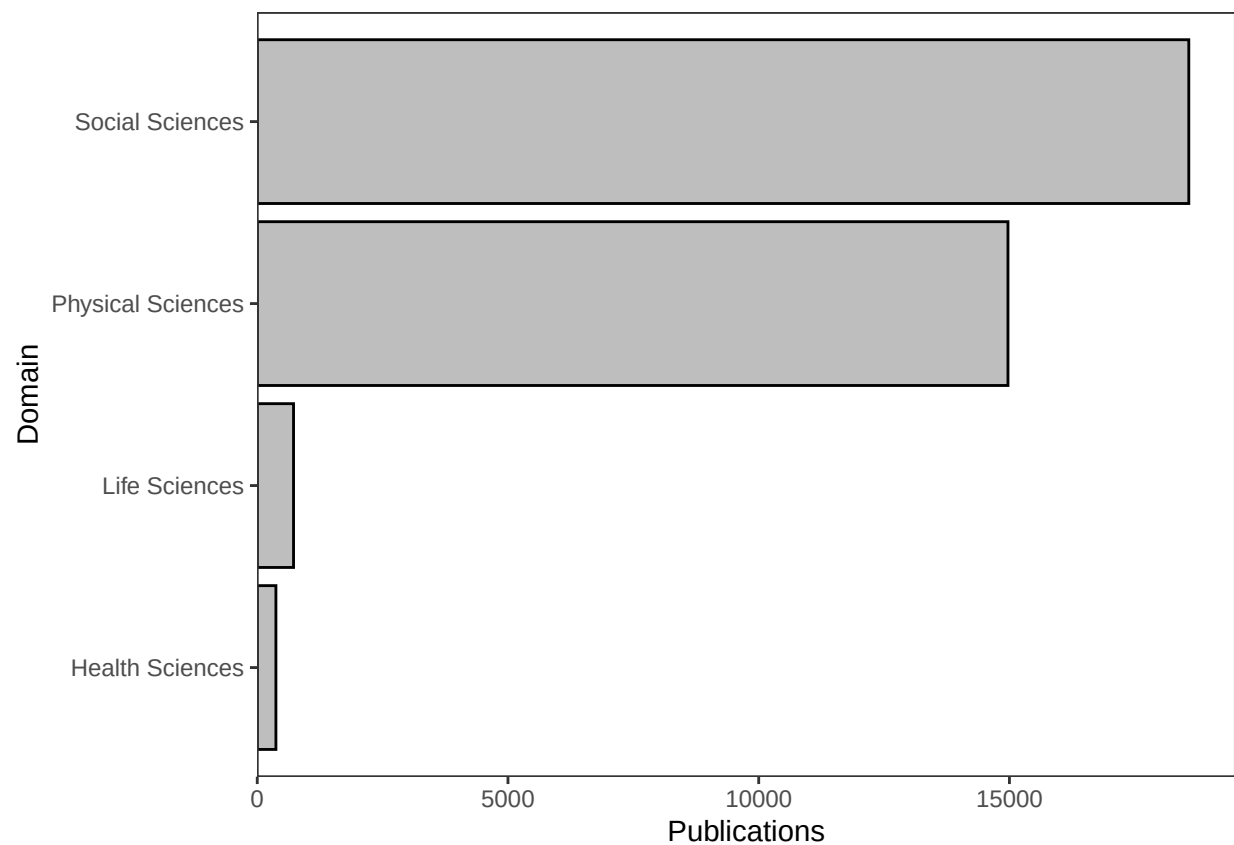


2. Publications over Time

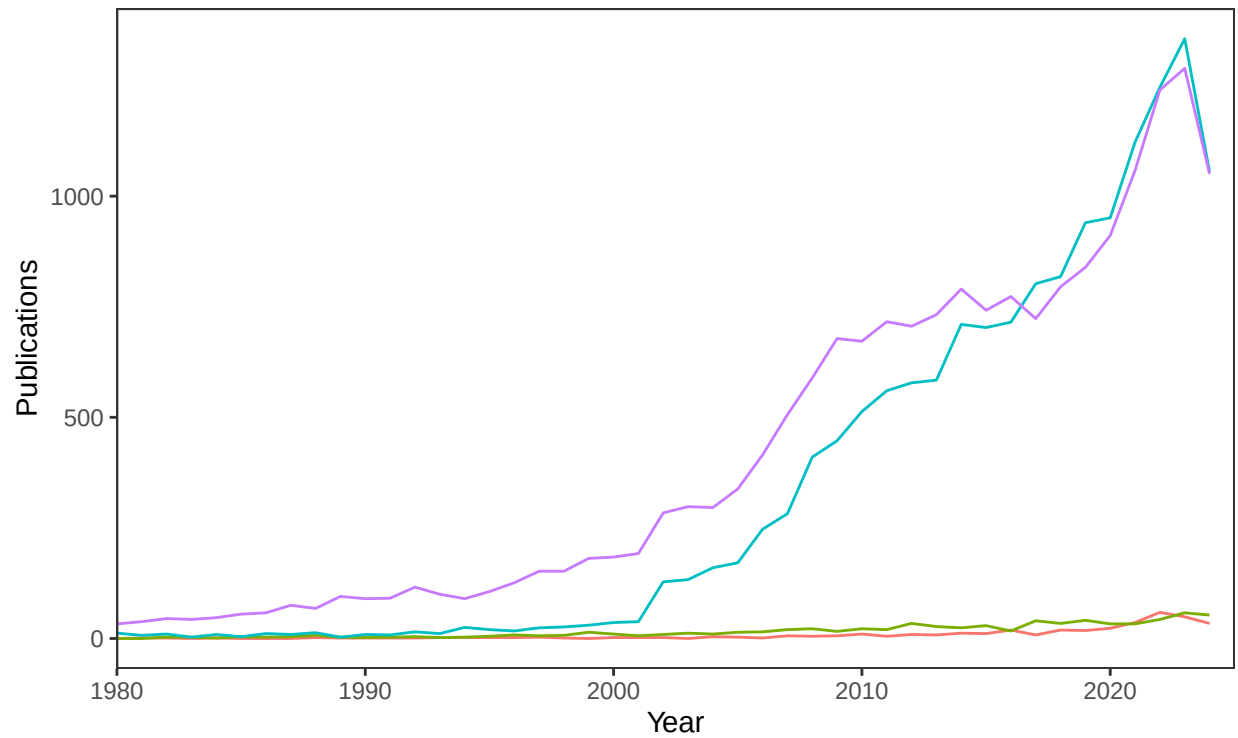


3. Domain Overview

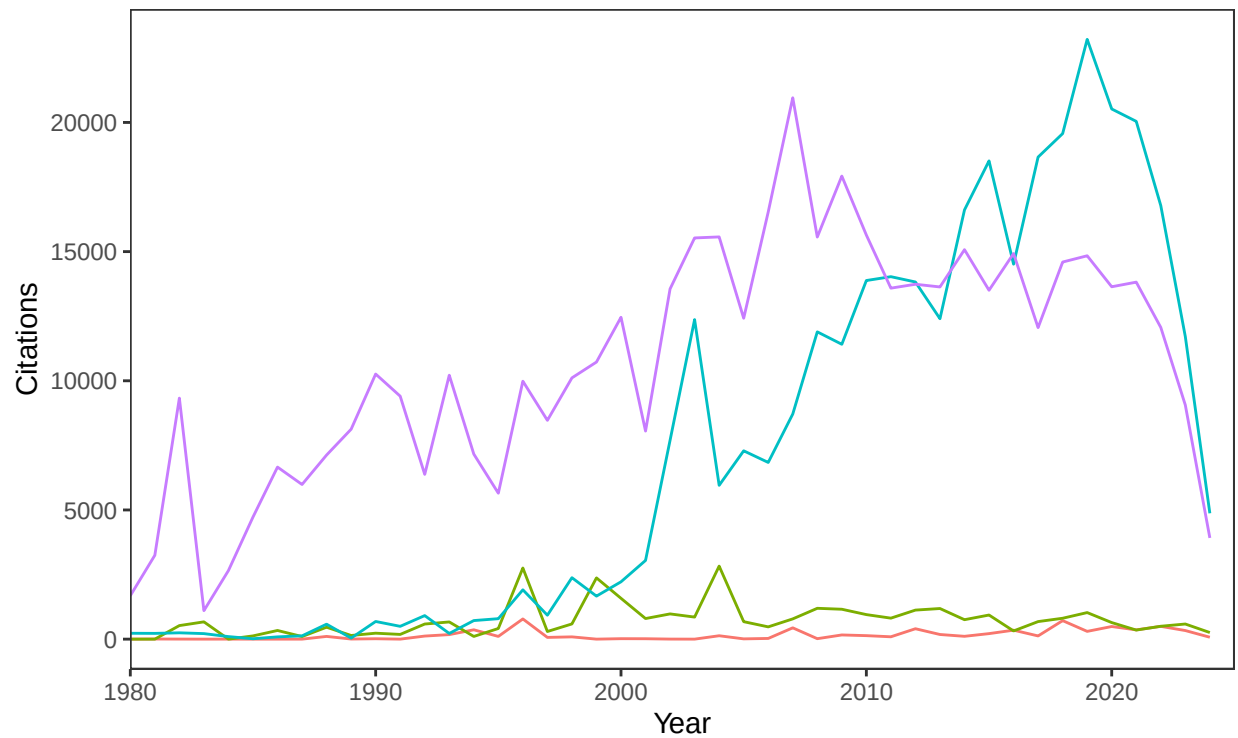
3.1 Top Domains



3.2 Domains over Time



Domain — Health Sciences — Life Sciences — Physical Sciences — Social Sciences



Domain — Health Sciences — Life Sciences — Physical Sciences — Social Sciences

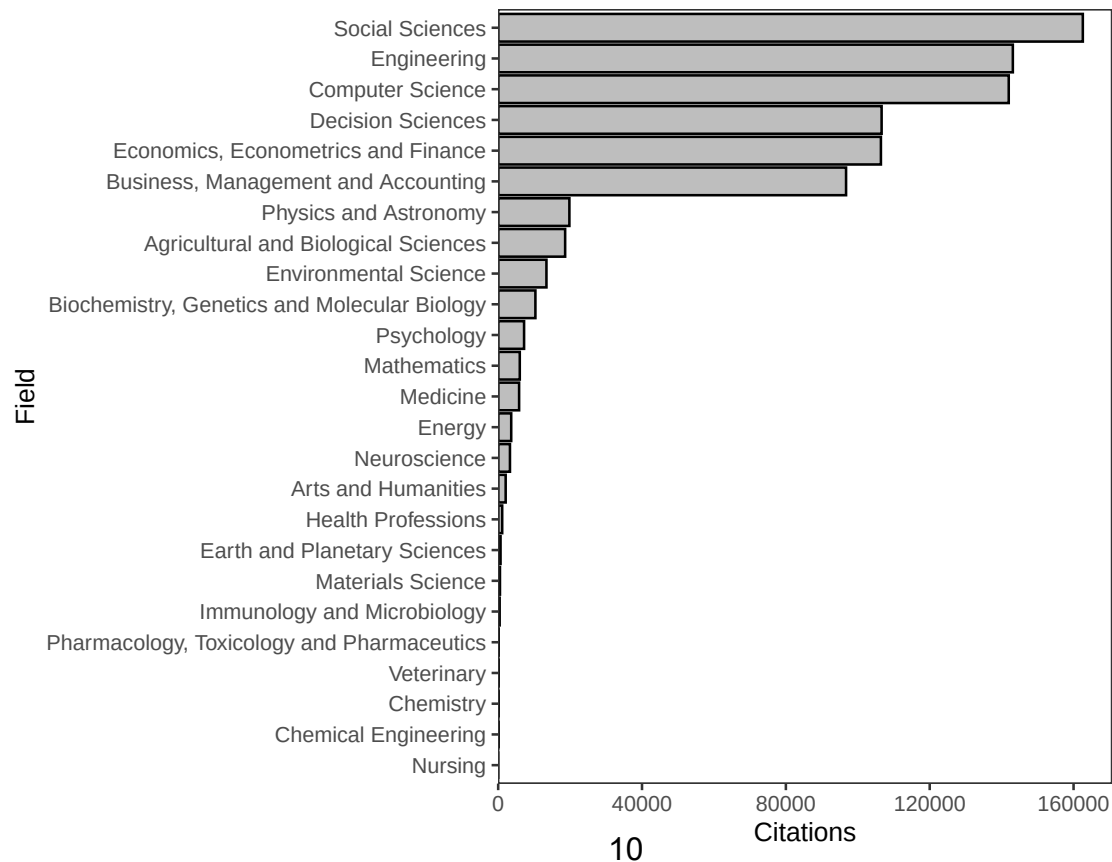
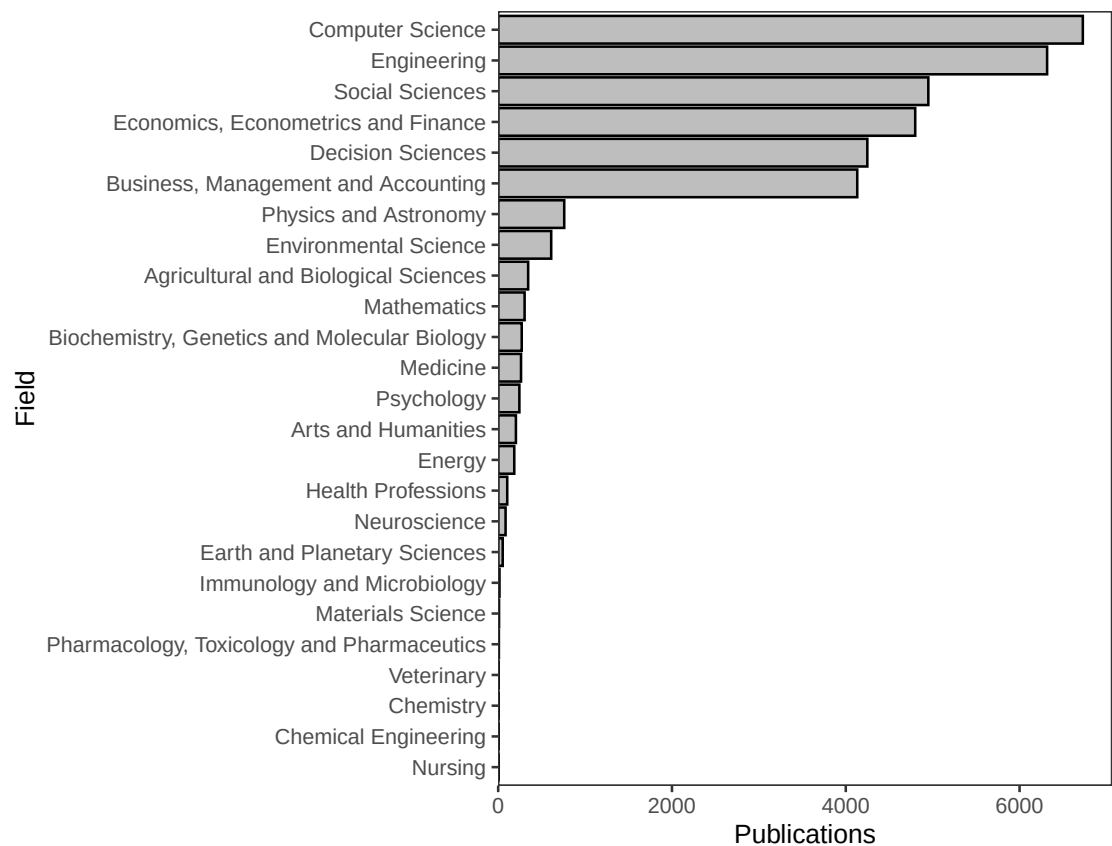
4. Field Overview

4.1 List of Fields

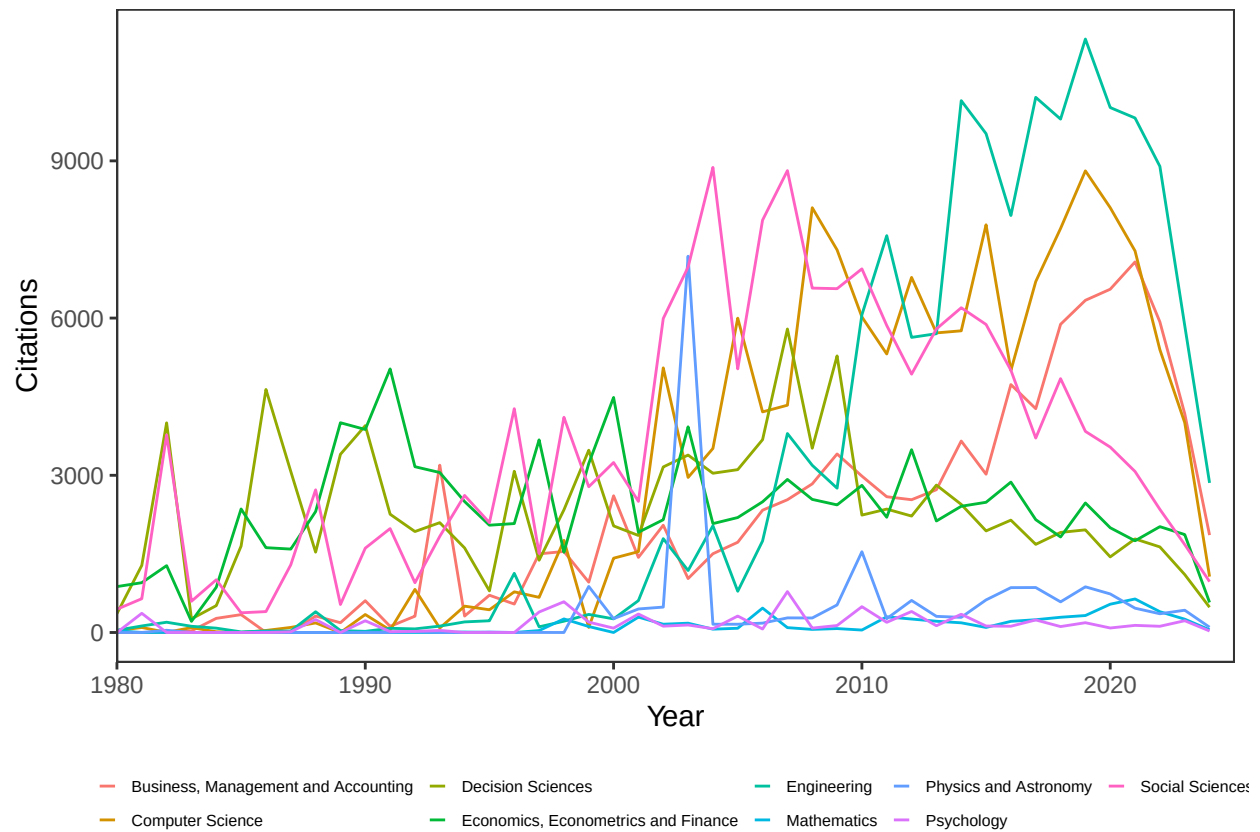
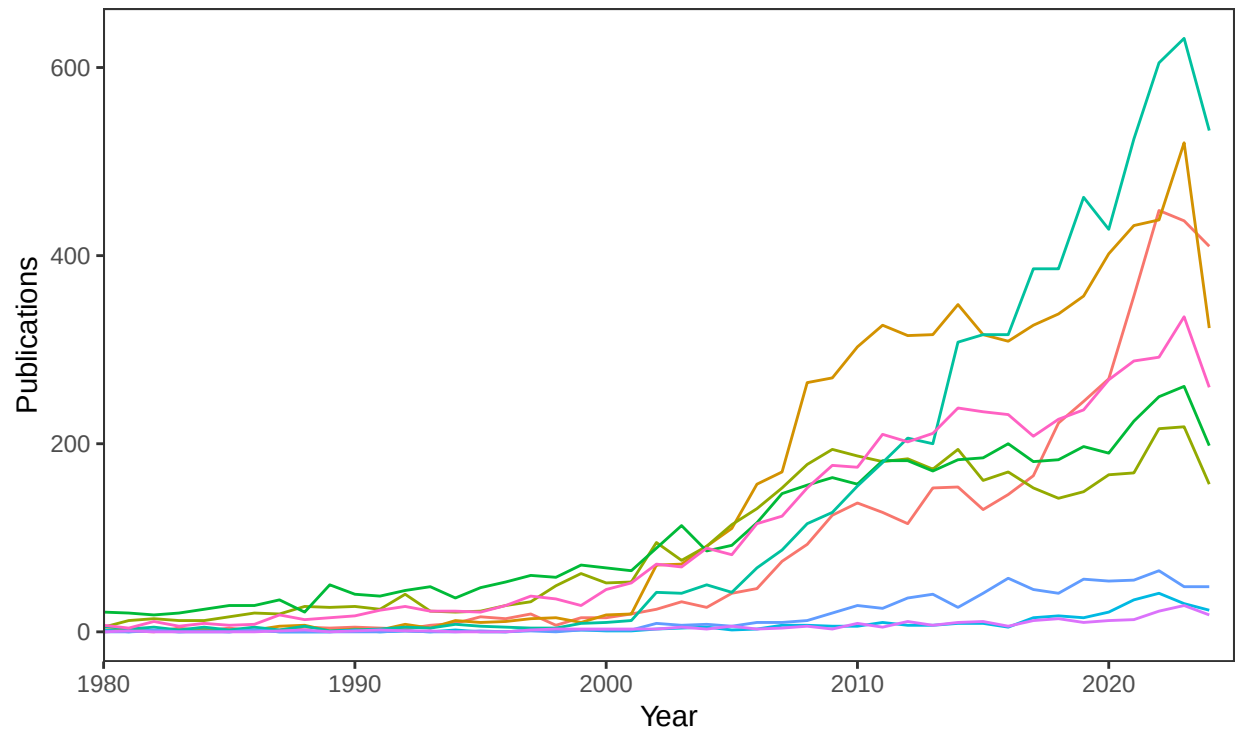
Table 4: List of Fields by OpenAlex.

Field
Agricultural and Biological Sciences
Arts and Humanities
Biochemistry, Genetics and Molecular Biology
Business, Management and Accounting
Chemical Engineering
Chemistry
Computer Science
Decision Sciences
Earth and Planetary Sciences
Economics, Econometrics and Finance
Energy
Engineering
Environmental Science
Health Professions
Immunology and Microbiology
Materials Science
Mathematics
Medicine
Neuroscience
Nursing
Pharmacology, Toxicology and Pharmaceutics
Physics and Astronomy
Psychology
Social Sciences
Veterinary

4.2 Top Fields



4.3 Fields over Time



4.4 Distribution of Fields across Queries

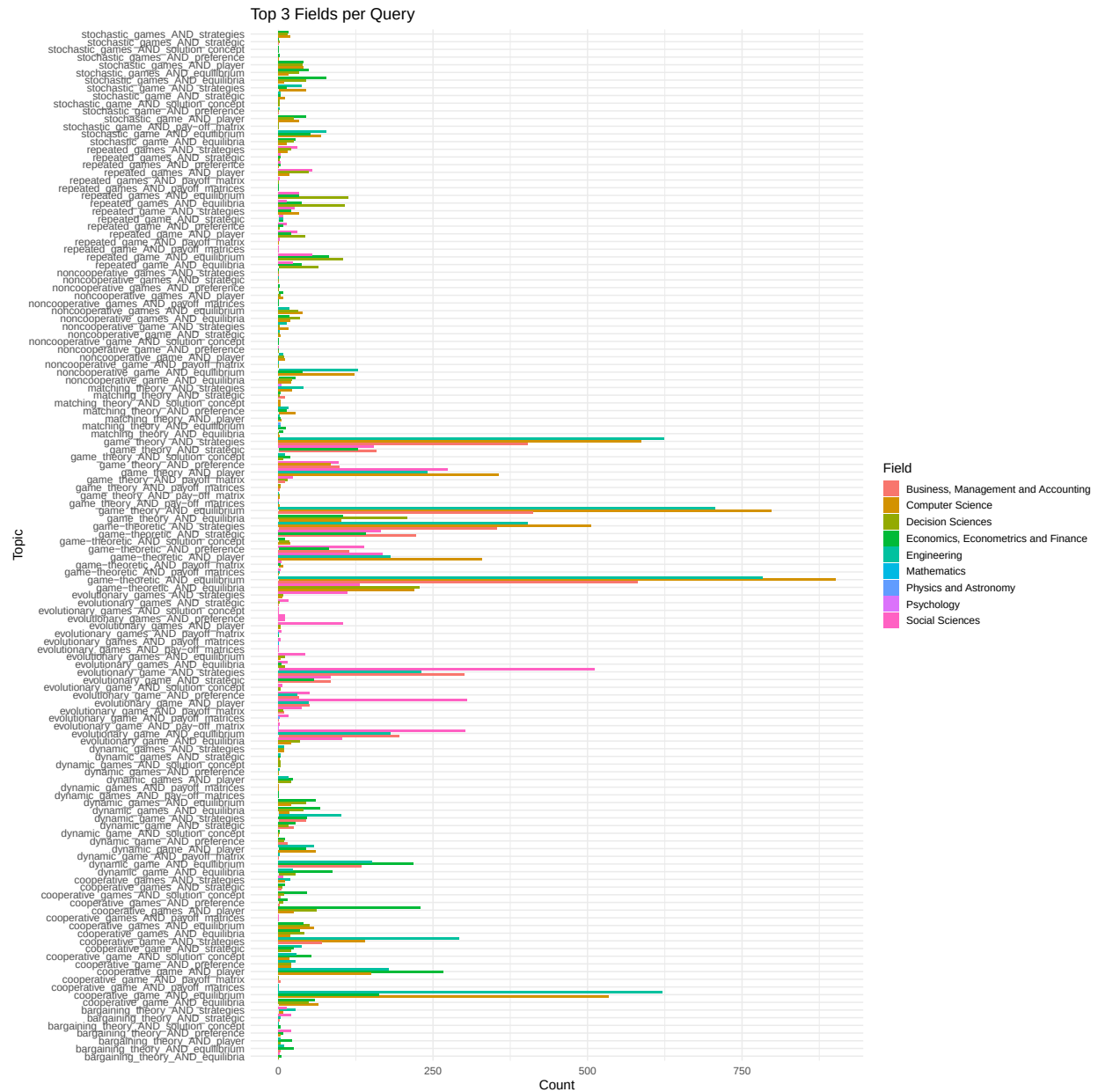
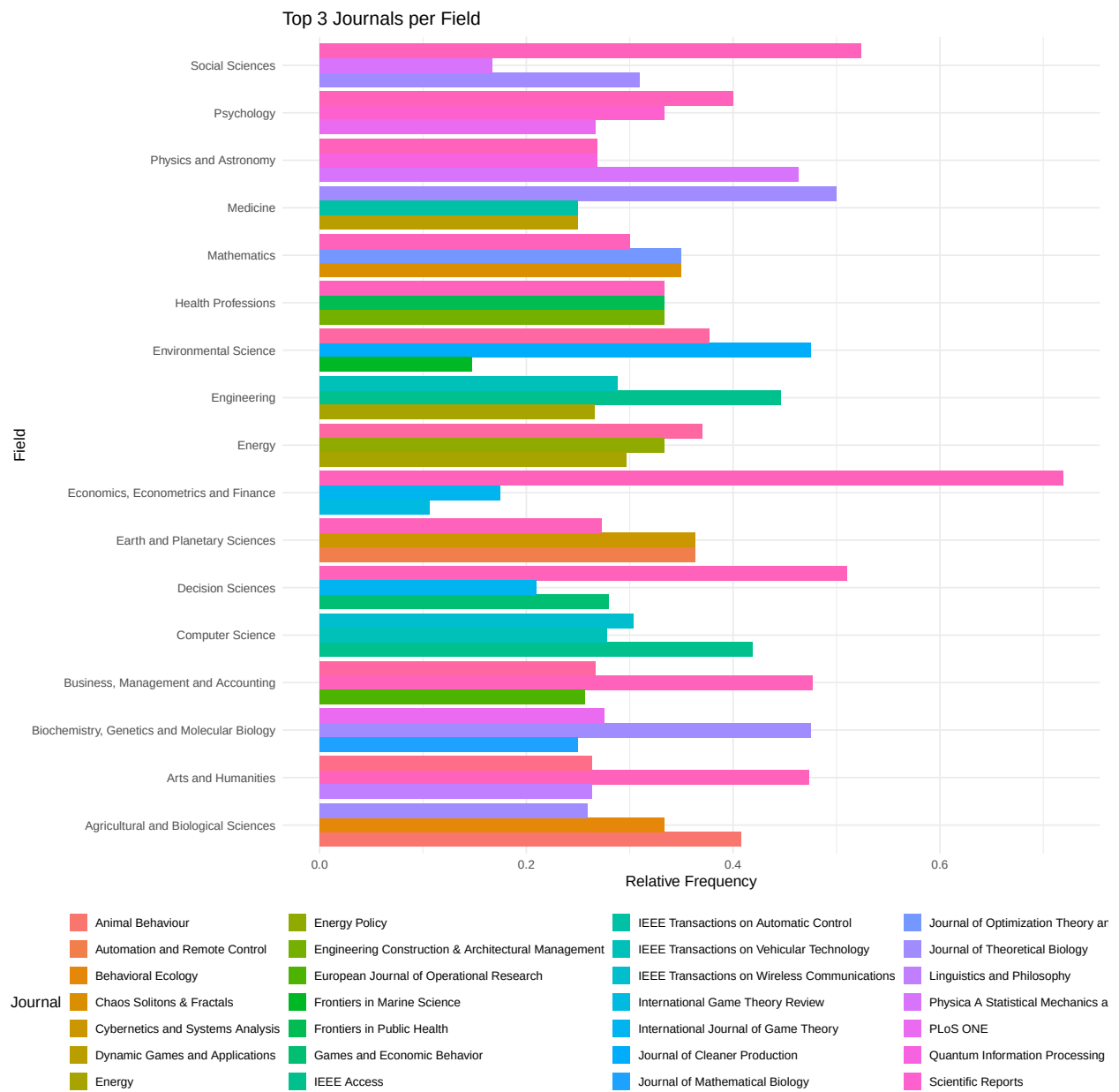


Table 5: Top Journals per Field

Field	Top Journals per Field (Descending Order)		
	Journal 1	Journal 2	Journal 3
Agricultural and Biological Sciences	Animal Behaviour (22)	Behavioral Ecology (18)	Journal of Theoretical Biology (14)
Arts and Humanities	SSRN Electronic Journal (9)	Linguistics and Philosophy (5)	Synthese (5)
Biochemistry, Genetics and Molecular Biology	Journal of Theoretical Biology (19)	PLoS ONE (11)	Journal of Mathematical Biology (10)
Business, Management and Accounting	SSRN Electronic Journal (225)	Sustainability (126)	European Journal of Operational Research (121)
Computer Science	IEEE Access (113)	IEEE Transactions on Wireless Communications (82)	IEEE Transactions on Vehicular Technology (75)
Decision Sciences	SSRN Electronic Journal (226)	Games and Economic Behavior (124)	International Journal of Game Theory (93)
Earth and Planetary Sciences	Automation and Remote Control (4)	Cybernetics and Systems Analysis (4)	SSRN Electronic Journal (3)
Economics, Econometrics and Finance	SSRN Electronic Journal (445)	International Journal of Game Theory (108)	International Game Theory Review (66)
Energy Engineering	Sustainability (10)	Energy Policy (9)	Energy (8)
	IEEE Access (141)	IEEE Transactions on Vehicular Technology (91)	Energy (84)
Environmental Science	Journal of Cleaner Production (29)	Sustainability (23)	Frontiers in Marine Science (9)
Health Professions	Engineering Construction & Architectural Management (4)	Frontiers in Public Health (4)	SSRN Electronic Journal (4)
Mathematics	Chaos Solitons & Fractals (7)	Journal of Optimization Theory and Applications (7)	SSRN Electronic Journal (6)
Medicine	Journal of Theoretical Biology (10)	Dynamic Games and Applications (5)	IEEE Transactions on Automatic Control (5)
Physics and Astronomy	Physica A Statistical Mechanics and its Applications (31)	Quantum Information Processing (18)	SSRN Electronic Journal (18)
Psychology	SSRN Electronic Journal (6)	Scientific Reports (5)	PLoS ONE (4)
Social Sciences	SSRN Electronic Journal (257)	Journal of Theoretical Biology (152)	Physica A Statistical Mechanics and its Applications (82)

4.5 Distribution of Journals across fields



4.6 Top Articles per Field

Table 6: Top Articles per Field

Field	Top Articles per Field (Descending Order)		
	Authors	Title	No. Citations
Business, Management and Accounting	Arvind Parkhe	strategic alliance structuring: a game theoretic and transaction cost examination of interfirm cooperation.	2581
	Jan W. Rivkin	imitation of complex strategies	1292
	Rikard Larsson;Lars Bengtsson;Kristina Henriksson;Judith Sparks	the interorganizational learning dilemma: collective knowledge development in strategic alliances	1077
Computer Science	Xu Chen;Lei Jiao;Wenzhong Li;Xiaoming Fu	efficient multi-user computation offloading for mobile-edge cloud computing	2552
	Cem Saraydar;Narayan B. Mandayam;D.J. Goodman	efficient power control via pricing in wireless data networks	1332
	Caroline Claus;Craig Boutilier	the dynamics of reinforcement learning in cooperative multiagent systems	1090
Decision Sciences	David M. Kreps;Robert Wilson	reputation and imperfect information	3101
	Drew Fudenberg;Eric Maskin	the folk theorem in repeated games with discounting or with incomplete information	2371
	Paul Milgrom;John Roberts	rationalizability, learning, and equilibrium in games with strategic complementarities	1816
Economics, Econometrics and Finance	Charles F. Manski	economic analysis of social interactions	2052
	Daniel Friedman	evolutionary games in economics	1797
	Barry R. Weingast	the political foundations of democracy and the rule of the law	1739

Engineering	Amir-Hamed Mohsenian-Rad;Vincent W. S. Wong;Juri Jatskevich;Robert Schober;Alberto Leon-Garcia	autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid	2711
	Amitav Mukherjee;S. Ali. A. Fakoorian;Jing Huang;A. Lee Swindlehurst	principles of physical layer security in multiuser wireless networks: a survey	1427
	Tamer Başar;P. Bernhard	h/sup ∞ /-Optimal control and related minimax design problems: a dynamic game approach	1066
Mathematics	Robert Axelrod;David Axelrod;Kenneth J. Pienta	evolution of cooperation among tumor cells	451
	Christopher A. Klausmeier;Elena Litchman	algal games: the vertical distribution of phytoplankton in poorly mixed water columns	294
	Donald A. Martin	the determinacy of blackwell games	259
Physics and Astronomy	David Kempe;Jon Kleinberg;Éva Tardos	maximizing the spread of influence through a social network	7088
	David Meyer	quantum strategies	744
	Andrea Montanari;Amin Saberi	the spread of innovations in social networks	384
Psychology	Herbert Gintis	a framework for the unification of the behavioral sciences	411
	Geoffrey F. Miller;Peter M. Todd	mate choice turns cognitive	368
	Calvin B. Harley	learning the evolutionarily stable strategy	366
Social Sciences	György Szabó;Gábor Fáth	evolutionary games on graphs	2823
	David M. Kreps;Paul Milgrom;John Roberts;Robert Wilson	rational cooperation in the finitely repeated prisoners' dilemma	2512
	Hisashi Ohtsuki;Christoph Hauert;E. James Lieberman;Martin A. Nowak	a simple rule for the evolution of cooperation on graphs and social networks	2155

5. Subfield Overview

5.1 List of Subfields

Table 7: List of Subfields by OpenAlex.

Subfield	No. Publications	No. Citations
Accounting	199	2386
Aerospace Engineering	658	11495
Aging	3	48
Agronomy and Crop Science	10	243
Algebra and Number Theory	2	14
Analytical Chemistry	1	10
Anthropology	4	48
Applied Mathematics	13	228
Applied Microbiology and Biotechnology	2	45
Applied Psychology	5	52
Aquatic Science	1	25
Archeology	2	4
Architecture	5	47
Artificial Intelligence	1356	25844
Astronomy and Astrophysics	13	95
Atmospheric Science	5	21
Atomic and Molecular Physics, and Optics	126	3162
Automotive Engineering	274	4685
Biomedical Engineering	75	1360
Biophysics	2	7
Building and Construction	187	4258
Business and International Management	41	459
Cancer Research	5	171
Cardiology and Cardiovascular Medicine	3	40
Cell Biology	2	57
Cellular and Molecular Neuroscience	2	166
Civil and Structural Engineering	243	6352
Classics	1	0
Clinical Psychology	28	517
Cognitive Neuroscience	80	3062
Communication	76	506
Complementary and alternative medicine	4	2
Computational Mathematics	9	425
Computational Mechanics	16	242
Computational Theory and Mathematics	490	10232
Computer Graphics and Computer-Aided Design	4	3

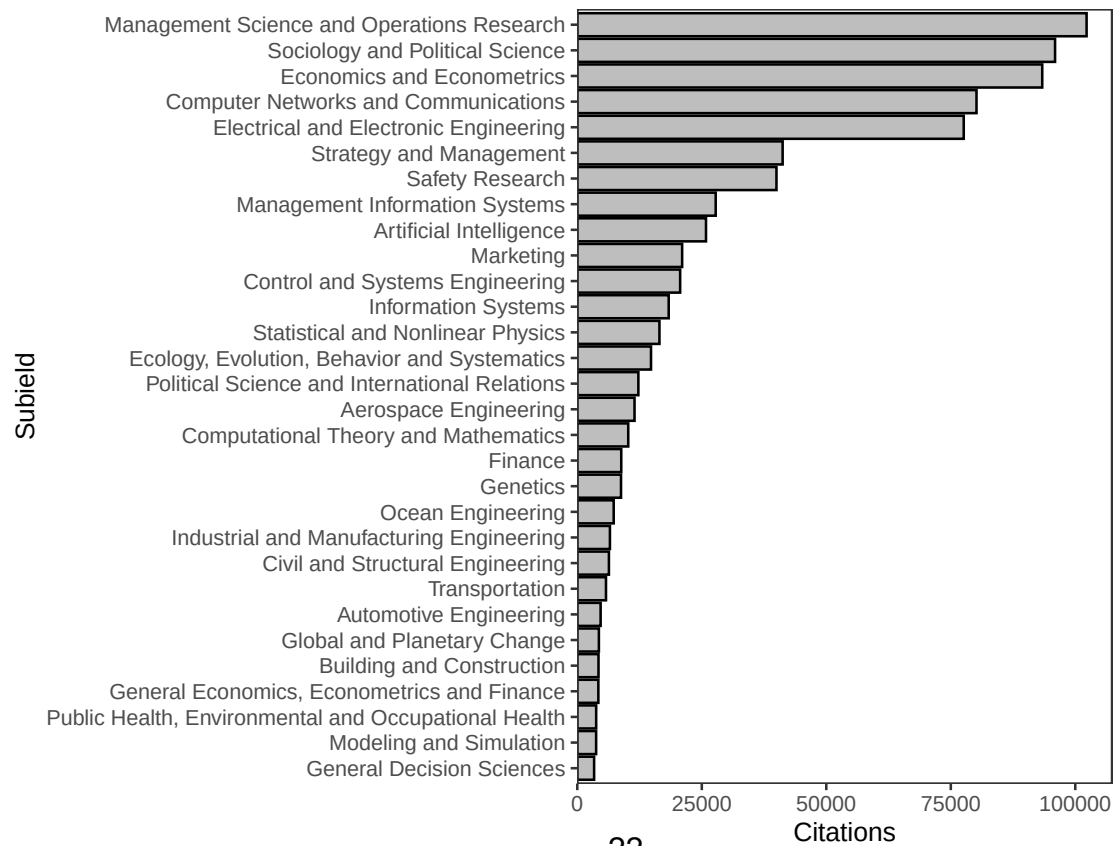
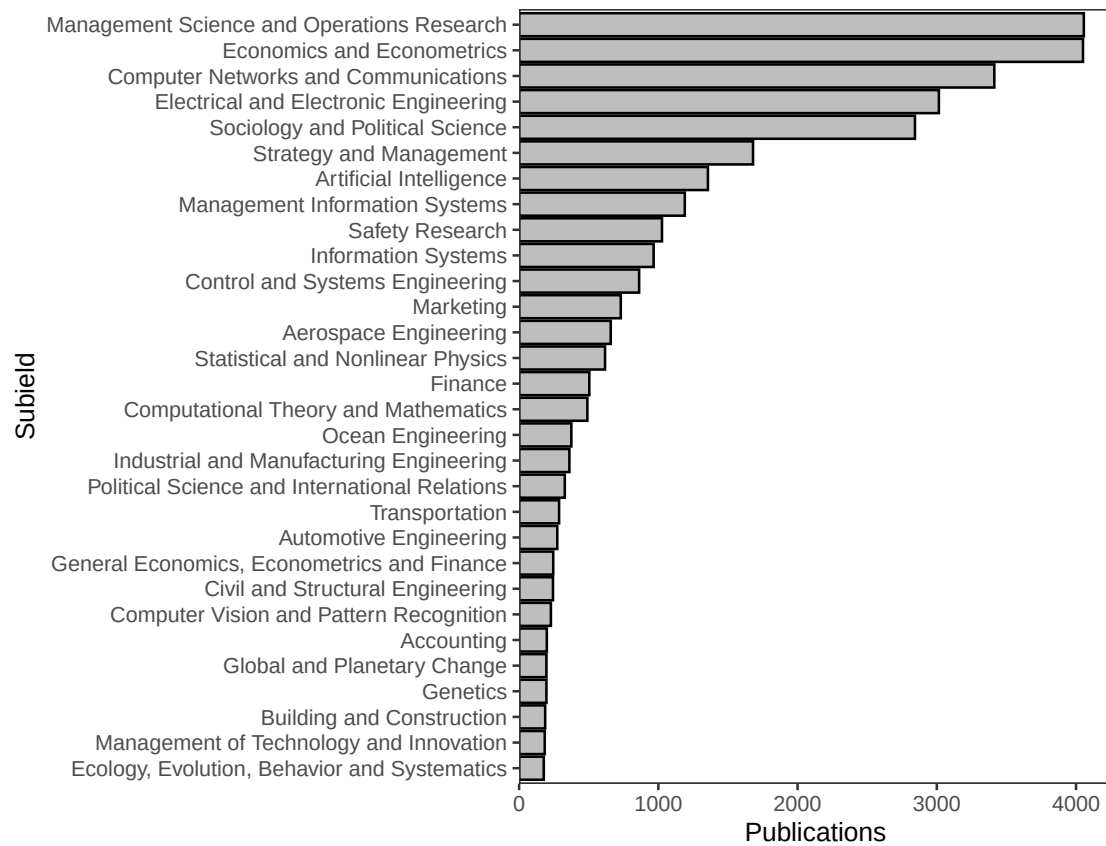
Computer Networks and Communications	3413	80165
Computer Science Applications	141	2931
Computer Vision and Pattern Recognition	228	2757
Control and Systems Engineering	862	20657
Cultural Studies	26	832
Demography	88	1683
Development	61	1158
Developmental and Educational Psychology	74	1854
Discrete Mathematics and Combinatorics	6	22
Earth-Surface Processes	39	154
Ecological Modeling	9	57
Ecology	50	1808
Ecology, Evolution, Behavior and Systematics	177	14803
Economics and Econometrics	4050	93366
Education	66	601
Electrical and Electronic Engineering	3015	77603
Electronic, Optical and Magnetic Materials	4	36
Emergency Medical Services	19	217
Emergency Medicine	2	53
Endocrinology	2	98
Endocrinology, Diabetes and Metabolism	1	4
Energy Engineering and Power Technology	27	175
Environmental Chemistry	4	20
Environmental Engineering	57	1448
Epidemiology	16	723
Experimental and Cognitive Psychology	52	1760
Family Practice	2	14
Finance	504	8827
Fluid Flow and Transfer Processes	2	16
Food Science	42	311
Fuel Technology	1	0
Gender Studies	50	2253
General Agricultural and Biological Sciences	31	222
General Decision Sciences	117	3380
General Economics, Econometrics and Finance	245	4231
General Energy	39	567
General Engineering	1	3
General Health Professions	27	121
General Psychology	1	224
General Social Sciences	6	31
Genetics	196	8782
Geography, Planning and Development	5	50
Geometry and Topology	26	389
Global and Planetary Change	196	4336
Hardware and Architecture	23	351

Health	11	62
Health Informatics	2	45
Health Information Management	4	15
Health, Toxicology and Mutagenesis	13	90
Hematology	1	6
Hepatology	2	35
History	3	111
History and Philosophy of Science	15	425
Human Factors and Ergonomics	1	2
Human-Computer Interaction	16	459
Immunology	8	56
Industrial and Manufacturing Engineering	361	6556
Infectious Diseases	13	258
Information Systems	966	18359
Information Systems and Management	30	206
Insect Science	3	24
Instrumentation	2	34
Language and Linguistics	22	265
Law	25	829
Library and Information Sciences	1	1
Linguistics and Language	7	186
Literature and Literary Theory	8	27
Management Information Systems	1190	27791
Management Science and Operations Research	4056	102270
Management of Technology and Innovation	184	2061
Management, Monitoring, Policy and Law	117	1214
Marketing	730	21054
Materials Chemistry	5	372
Mathematical Physics	10	137
Mechanical Engineering	56	505
Mechanics of Materials	15	256
Media Technology	146	1418
Microbiology	1	5
Modeling and Simulation	162	3783
Molecular Biology	62	1185
Museology	3	34
Music	2	3
Nature and Landscape Conservation	29	1887
Neurology	6	63
Neuropsychology and Physiological Psychology	1	2
Nuclear and High Energy Physics	2	14
Numerical Analysis	20	187
Nutrition and Dietetics	1	0
Obstetrics and Gynecology	1	30
Ocean Engineering	375	7309

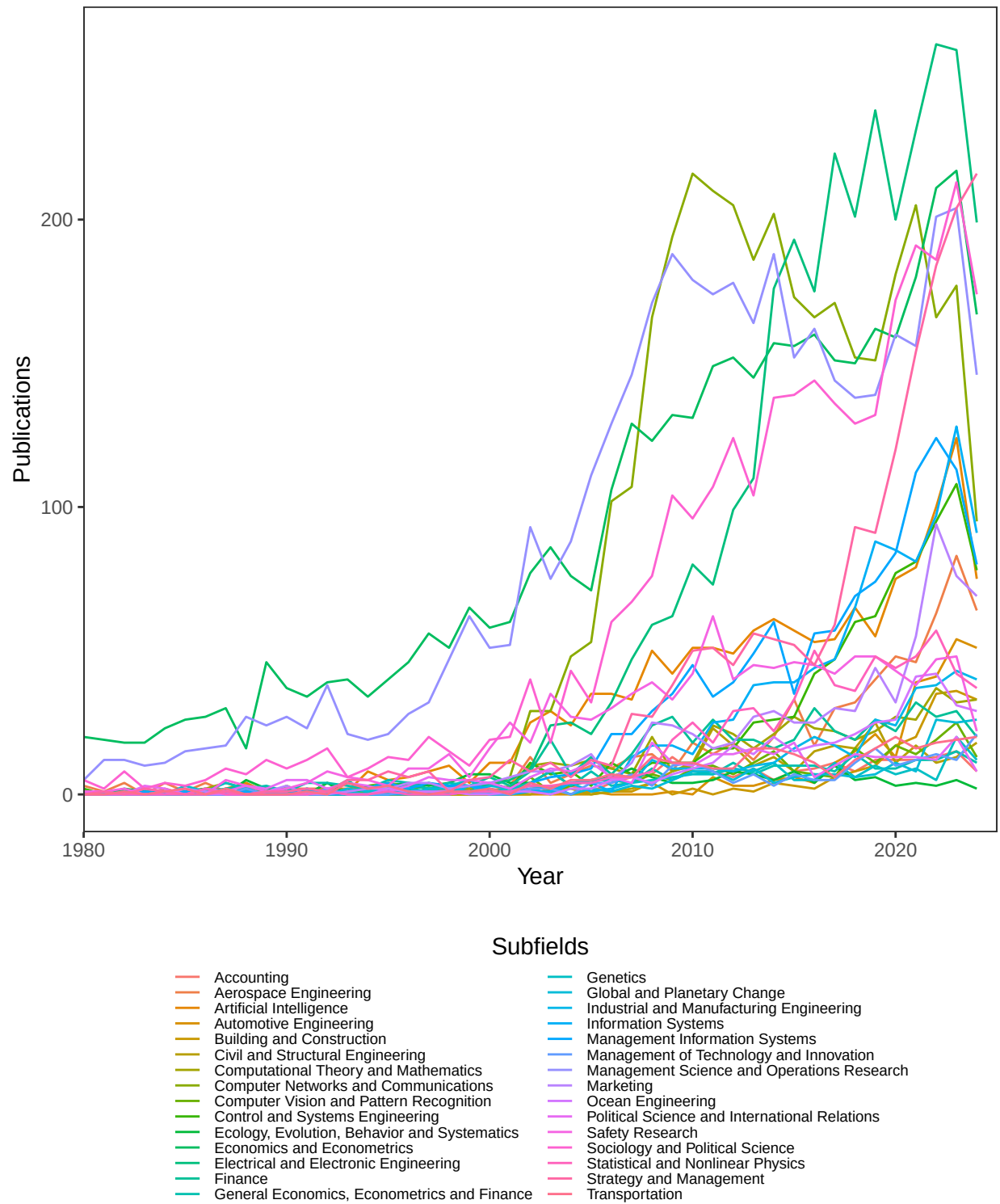
Oceanography	6	447
Oncology	2	28
Organizational Behavior and Human Resource Management	99	1685
Orthopedics and Sports Medicine	13	228
Paleontology	1	32
Parasitology	1	40
Pediatrics, Perinatology and Child Health	3	43
Pharmacology	10	129
Pharmacy	8	65
Philosophy	136	922
Physical Therapy, Sports Therapy and Rehabilitation	7	5
Physiology	4	17
Plant Science	36	792
Political Science and International Relations	328	12265
Pollution	29	668
Polymers and Plastics	1	85
Psychiatry and Mental health	3	15
Public Administration	19	147
Public Health, Environmental and Occupational Health	162	3791
Pulmonary and Respiratory Medicine	9	266
Radiological and Ultrasound Technology	39	693
Radiology, Nuclear Medicine and Imaging	6	76
Rehabilitation	4	80
Renewable Energy, Sustainability and the Environment	118	2882
Safety Research	1026	39994
Safety, Risk, Reliability and Quality	104	1775
Sensory Systems	1	16
Signal Processing	75	795
Small Animals	4	67
Social Psychology	83	2792
Sociology and Political Science	2843	95940
Software	16	76
Soil Science	45	2198
Space and Planetary Science	1	3
Spectroscopy	2	9
Statistical and Nonlinear Physics	617	16497
Statistics and Probability	55	821
Statistics, Probability and Uncertainty	45	767
Strategy and Management	1680	41233
Surgery	2	16
Theoretical Computer Science	1	0
Tourism, Leisure and Hospitality Management	9	66
Transplantation	1	4
Transportation	287	5763
Urban Studies	19	223

Virology	3	260
Visual Arts and Performing Arts	13	295
Water Science and Technology	30	486

5.2 Top Subfields



5.3 Subfields over Time



5.4 Distribution of Journals across Subfields

Table 8: Top Journals per Subfield

Subfield	Top Journals per Subfield (Descending Order)		
	Journal 1	Journal 2	Journal 3
Accounting	SSRN Electronic Journal (32)	Journal of Accounting and Public Policy (3)	Journal of Public Economics (3)
Aerospace Engineering	IEEE Access (24)	IEEE Transactions on Aerospace and Electronic Systems (17)	Journal of Guidance Control and Dynamics (16)
Artificial Intelligence	Adaptive Agents and Multi-Agents Systems (28)	Proceedings of the AAAI Conference on Artificial Intelligence (22)	IEEE Access (17)
Automotive Engineering	SSRN Electronic Journal (15)	IEEE Transactions on Intelligent Transportation Systems (10)	Transportation Research Part C Emerging Technologies (10)
Building and Construction	Sustainability (19)	Journal of Cleaner Production (13)	Building and Environment (7)
Civil and Structural Engineering	Reliability Engineering & System Safety (17)	European Journal of Operational Research (12)	Risk Analysis (12)
Computational Theory and Mathematics	Theoretical Computer Science (11)	Optimization (10)	IEEE Transactions on Cybernetics (8)
Computer Networks and Communications	IEEE Transactions on Wireless Communications (80)	IEEE Transactions on Vehicular Technology (66)	IEEE Access (56)
Computer Vision and Pattern Recognition	Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE (5)	IEEE Access (4)	IEEE Transactions on Circuits and Systems for Video Technology (4)
Control and Systems Engineering	IEEE Transactions on Automatic Control (23)	IEEE Transactions on Intelligent Transportation Systems (18)	Automatica (11)
Ecology, Evolution, Behavior and Systematics	Animal Behaviour (22)	Behavioral Ecology (18)	Journal of Theoretical Biology (14)
Economics and Econometrics	SSRN Electronic Journal (359)	International Journal of Game Theory (107)	International Game Theory Review (64)
Electrical and Electronic Engineering	IEEE Access (86)	Energy (75)	Applied Energy (65)
Finance	SSRN Electronic Journal (65)	SIAM Journal on Control and Optimization (9)	Dynamic Games and Applications (7)
General Economics, Econometrics and Finance	SSRN Electronic Journal (21)	IFAC Proceedings Volumes (5)	International Journal of Industrial Organization (3)
Genetics	Journal of Theoretical Biology (17)	Journal of Mathematical Biology (10)	The American Naturalist (10)
Global and Planetary Change	Sustainability (7)	Environmental and Resource Economics (6)	Fisheries Research (6)

Industrial and Manufacturing Engineering Information Systems	Journal of Cleaner Production (12) IEEE Access (27)	Sustainability (12) SSRN Electronic Journal (17)	Maritime Policy & Management (8) Mathematics (15)
Management Information Systems	European Journal of Operational Research (64)	SSRN Electronic Journal (49)	International Journal of Production Economics (46)
Management Science and Operations Research	SSRN Electronic Journal (213)	Games and Economic Behavior (123)	International Journal of Game Theory (91)
Management of Technology and Innovation	SSRN Electronic Journal (6)	Sustainability (5)	Applied Mathematics and Nonlinear Sciences (3)
Marketing	SSRN Electronic Journal (64)	Marketing Science (32)	Management Science (18)
Ocean Engineering	Water Resources Management (25)	Water (12)	Sustainability (9)
Political Science and International Relations	SSRN Electronic Journal (23)	Journal of Theoretical Politics (12)	American Political Science Review (8)
Safety Research	SSRN Electronic Journal (138)	Games and Economic Behavior (36)	Journal of Economic Behavior & Organization (35)
Sociology and Political Science	Journal of Theoretical Biology (145)	Physica A Statistical Mechanics and its Applications (77)	Applied Mathematics and Computation (68)
Statistical and Nonlinear Physics	Physica A Statistical Mechanics and its Applications (18)	SSRN Electronic Journal (17)	Complexity (10)
Strategy and Management	Sustainability (77)	SSRN Electronic Journal (66)	Journal of Cleaner Production (46)
Transportation	Transportation Research Part B Methodological (17)	Transportation Research Record Journal of the Transportation Research Board (8)	IFAC Proceedings Volumes (7)



5.5 Top Articles per Subfield

Table 9: Top Articles per subfield

Top Articles per Subfield (Descending Order)			
Subfield	Authors	Title	No. Citations
Aerospace Engineering	Jingjing Cui;Yuanwei Liu;Arumugam Nallanathan	multi-agent reinforcement learning-based resource allocation for uav networks	498
	Mahmut Parlar	game theoretic analysis of the substitutable product inventory problem with random demands	352
	Liang Xiao;Xiaozhen Lu;Dongjin Xu;Yuliang Tang;Lei Wang;Weihua Zhuang	uav relay in vanets against smart jamming with reinforcement learning	292
Artificial Intelligence	Xu Chen;Lei Jiao;Wenzhong Li;Xiaoming Fu	efficient multi-user computation offloading for mobile-edge cloud computing	2552
	Caroline Claus;Craig Boutilier	the dynamics of reinforcement learning in cooperative multiagent systems	1090
	Marc Pauly	a modal logic for coalitional power in games	466
Automotive Engineering	Adam Millard□Ball	pedestrians, autonomous vehicles, and cities	291
	Xiaoxiang Na;David J. Cole	game-theoretic modeling of the steering interaction between a human driver and a vehicle collision avoidance controller	188
	Zhu Xiao;Xingxia Dai;Hongbo Jiang;Dong Wang;Hongyang Chen;Liang Yang;Fanzi Zeng	vehicular task offloading via heat-aware mec cooperation using game-theoretic method	168
Building and Construction	Lei Du;Yingbin Feng;Wei Lu;Lingkai Kong;Zhi Yang	evolutionary game analysis of stakeholders' decision-making behaviours in construction and demolition waste management	148

Civil and Structural Engineering	Qinghua He;Zilun Wang;Ge Wang;Jian Zuo;Guangdong Wu;Bingsheng Liu	to be green or not to be: how environmental regulations shape contractor greenwashing behaviors in construction projects	131
	Ke Fan;Eddie C.M. Hui	evolutionary game theory analysis for understanding the decision-making mechanisms of governments and developers on green building incentives	126
	Jun Zhuang;Vicki M. Bier;Oğuzhan Alagöz	modeling secrecy and deception in a multiple-period attacker–defender signaling game	268
	Eric Shieh;Bo An;Rong Yang;Milind Tambe;Craig Baldwin;Joseph DiRenzo;Ben Maule;G. Dale Meyer	protect: a deployed game theoretic system to protect the ports of the united states	208
Computational Theory and Mathematics	Vicki M. Bier;Aniruddha Nagaraj;Vinod Abhichandani	protection of simple series and parallel systems with components of different values	204
	Anne Condon	the complexity of stochastic games	501
	Sjur Didrik Flåm;A. C. Антипин	equilibrium programming using proximal-like algorithms	437
Computer Networks and Communications	James T. Moore;Jonathan F. Bard	the mixed integer linear bilevel programming problem	343
	Cem Saraydar;Narayan B. Mandayam;D.J. Goodman	efficient power control via pricing in wireless data networks	1332
	Xu Chen;Xu Chen	decentralized computation offloading game for mobile cloud computing	843
Control and Systems Engineering	D.J. Goodman;Narayan B. Mandayam	power control for wireless data	652
	Tamer Başar;P. Bernhard	$h/\sup \infty/-$ Optimal control and related minimax design problems: a dynamic game approach	1066

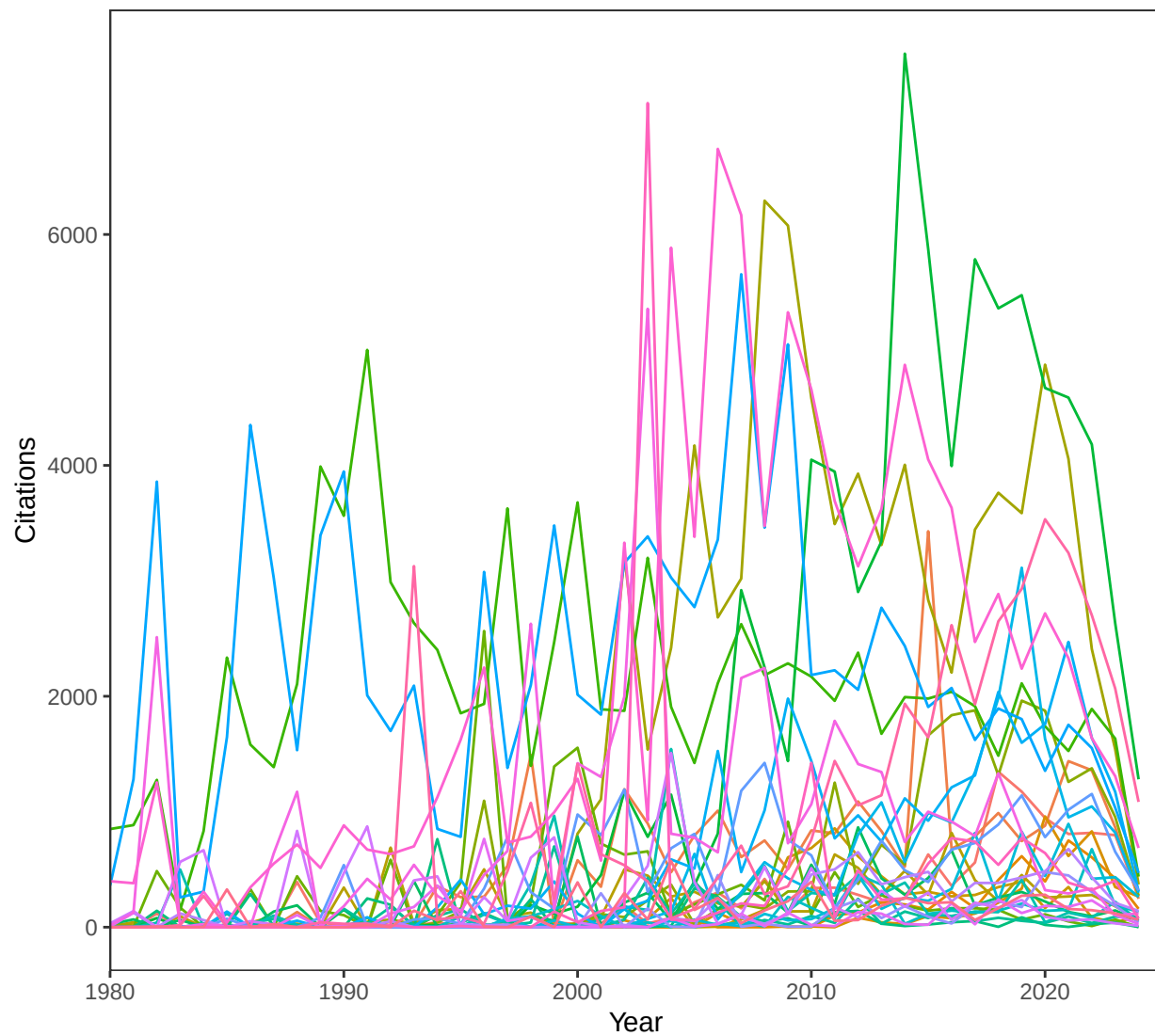
Ecology, Evolution, Behavior and Systematics	Yang Han;Ke Zhang;Hong Li;E.A.A. Coelho;Josep M. Guerrero	mas-based distributed coordinated control and optimization in microgrid and microgrid clusters: a comprehensive overview	455
	Hao Wang;Jianwei Huang;Hao Wang;Jianwei Huang	incentivizing energy trading for interconnected microgrids	447
	Mart R. Gross	alternative reproductive strategies and tactics: diversity within sexes	1681
	Hanna Kokko	competition for early arrival in migratory birds	812
	Stephen C. Stearns	life history evolution: successes, limitations, and prospects	793
Economics and Econometrics	Charles F. Manski	economic analysis of social interactions	2052
	Daniel Friedman	evolutionary games in economics	1797
	Barry R. Weingast	the political foundations of democracy and the rule of the law	1739
Electrical and Electronic Engineering	Amir-Hamed Mohsenian-Rad;Vincent W. S. Wong;Juri Jatskevich;Robert Schober;Alberto Leon-Garcia	autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid	2711
	Amitav Mukherjee;S. Ali. A. Fakoorian;Jing Huang;A. Lee Swindlehurst	principles of physical layer security in multiuser wireless networks: a survey	1427
	Zhongjing Ma;Duncan S. Callaway;Ian A. Hiskens	decentralized charging control of large populations of plug-in electric vehicles	963
Finance	Tomas Björk;Agatha Murgoci;Xun Yu Zhou	mean–variance portfolio optimization with state-dependent risk aversion	508
	Hua Fan;Suresh Sundaresan	debt valuation, renegotiation, and optimal dividend policy	444
	Tomas Björk;Agatha Murgoci	a general theory of markovian time inconsistent stochastic control problems	340
General Decision Sciences	Colin F. Camerer;Teck-Hua Ho	violations of the betweenness axiom and nonlinearity in probability	740
	Colin F. Camerer;George Loewenstein;Dražen Prelec	neuroeconomics: how neuroscience can inform economics	303

General Economics, Econometrics and Finance	Teck-Hua Ho;Noah Lim;Colin F. Camerer	modeling the psychology of consumer and firm behavior with behavioral economics	262
	William A. Bomberger;Thomas J. Sargent	the conquest of american inflation	673
	Thad Dunning	resource dependence, economic performance, and political stability	265
	Francisco J. Ruge-Murciá	inflation targeting under asymmetric preferences	163
Genetics	Martin A. Nowak;Karl Sigmund	evolutionary dynamics of biological games	1140
	Paul E. Turner;Lin Chao	prisoner's dilemma in an rna virus	659
Global and Planetary Change	Hans J. Bremermann;John Pickering	a game-theoretical model of parasite virulence	387
	Oskar Franklin;Jacob Johansson;Roderick C. Dewar;Ulf Dieckmann;R. E. McMurtrie;Åke Brännström;Ray Dybzinski	modeling carbon allocation in trees: a search for principles	291
	Jiaqi Peng;Jianmin Zhang	urban flooding risk assessment based on gis- game theory combination weight: a case study of zhengzhou city	169
	Scott Barrett;Astrid Dannenberg	sensitivity of collective action to uncertainty about climate tipping points	168
Industrial and Manufacturing Engineering	Yuandou Wang;Hang Liu;Wanbo Zheng;Yunni Xia;Yawen Li;Peng Chen;Kunyin Guo;Hong Xie	multi-objective workflow scheduling with deep-q-network-based multi-agent reinforcement learning	271
	Leonard Heilig;Eduardo Lalla-Ruiz;Stefan Voß	digital transformation in maritime ports: analysis and a game theoretic framework	251
Information Systems	Masahiro Ishii;Paul Tae-Woo Lee;Koichiro Tezuka;Young-Tae Chang	a game theoretical analysis of port competition	189
	Wenbo Wang;Dinh Thai Hoang;Peizhao Hu;Zehui Xiong;Dusit Niyato;Ping Wang;Yonggang Wen;Dong In Kim	a survey on consensus mechanisms and mining strategy management in blockchain networks	940

Management Information Systems	David Easley;Maureen O'Hara;Soumya Basu	from mining to markets: the evolution of bitcoin transaction fees	547
	Kong-wei Lye;Jeannette M. Wing	game strategies in network security	433
	Mahesh Nagarajan;Greys Sošić	game-theoretic analysis of cooperation among supply chain agents: review and extensions	607
	Jan A. Van Mieghem	coordinating investment, production, and subcontracting	361
	Gary H. Chao;Seyed M. R. Iravani;R. Canan Savaskan	quality improvement incentives and product recall cost sharing contracts	358
Management Science and Operations Research	David M. Kreps;Robert Wilson	reputation and imperfect information	3101
	Drew Fudenberg;Eric Maskin	the folk theorem in repeated games with discounting or with incomplete information	2371
	Paul Milgrom;John Roberts	rationalizability, learning, and equilibrium in games with strategic complementarities	1816
Marketing	Patrick Bajari;C. Lanier Benkard;Jonathan Levin	estimating dynamic models of imperfect competition	869
	Marcel Corstjens;Rajiv Lal	building store loyalty through store brands	667
	Michael J. Mazzeo	product choice and oligopoly market structure	634
Modeling and Simulation	Robert Axelrod;David Axelrod;Kenneth J. Pienta	evolution of cooperation among tumor cells	451
	Christopher A. Klausmeier;Elena Litchman	algal games: the vertical distribution of phytoplankton in poorly mixed water columns	294
	Sheryl L. Chang;Mahendra Piraveenan;Philippa Pattison;Mikhail Prokopenko	game theoretic modelling of infectious disease dynamics and intervention methods: a review	159
Ocean Engineering	Kaveh Madani;Jay R. Lund	a monte-carlo game theoretic approach for multi-criteria decision making under uncertainty	188
	Kaveh Madani;Keith W. Hipel	non-cooperative stability definitions for strategic analysis of generic water resources conflicts	148

	Lu Xiao;Jianyue Liu;Jinwen Ge	dynamic game in agriculture and industry cross-sectoral water pollution governance in developing countries	144
Political Science and International Relations	Alberto Alesina	credibility and policy convergence in a two-party system with rational voters	823
	Wolfgang Mayer	endogenous tariff formation.	666
	Vivien A. Schmidt;Claudio M. Radaelli	policy change and discourse in europe: conceptual and methodological issues	605
Public Health, Environmental and Occupational Health	Gerry Mackie	ending footbinding and infibulation: a convention account	765
	Don M. Hugie;Lawrence M. Dill	fish and game: a game theoretic approach to habitat selection by predators and prey*	260
	Arja Kaitala;Veijo Kaitala;Per Lundberg	a theory of partial migration	172
Safety Research	David M. Kreps;Paul Milgrom;John Roberts;Robert Wilson	rational cooperation in the finitely repeated prisoners' dilemma	2512
	Martin Dufwenberg;Georg Kirchsteiger	a theory of sequential reciprocity	1902
	Ido Erev;Alvin E. Roth	predicting how people play games: reinforcement learning in experimental games with unique, mixed strategy equilibria	1822
Sociology and Political Science	György Szabó;Gábor Fáth	evolutionary games on graphs	2823
	Hisashi Ohtsuki;Christoph Hauert;E. James Lieberman;Martin A. Nowak	a simple rule for the evolution of cooperation on graphs and social networks	2155
	Martin A. Nowak;Akira Sasaki;Christine Taylor;Drew Fudenberg	emergence of cooperation and evolutionary stability in finite populations	1378
Statistical and Nonlinear Physics	David Kempe;Jon Kleinberg;Éva Tardos	maximizing the spread of influence through a social network	7088
	Andrea Montanari;Amin Saberi	the spread of innovations in social networks	384

Strategy and Management	Ramasuri Narayanam;Y. Narahari	a shapley value-based approach to discover influential nodes in social networks	373
	Arvind Parkhe	strategic alliance structuring: a game theoretic and transaction cost examination of interfirm cooperation.	2581
	Jan W. Rivkin	imitation of complex strategies	1292
	Rikard Larsson;Lars Bengtsson;Kristina Henriksson;Judith Sparks	the interorganizational learning dilemma: collective knowledge development in strategic alliances	1077
Transportation	José Correa;Andreas S. Schulz;Nicolás E. Stier-Moses	selfish routing in capacitated networks	386
	Michael G.H. Bell	a game theory approach to measuring the performance reliability of transport networks	379
	Alain Haurie;Patrice Marcotte	on the relationship between nash—cournot and wardrop equilibria	327

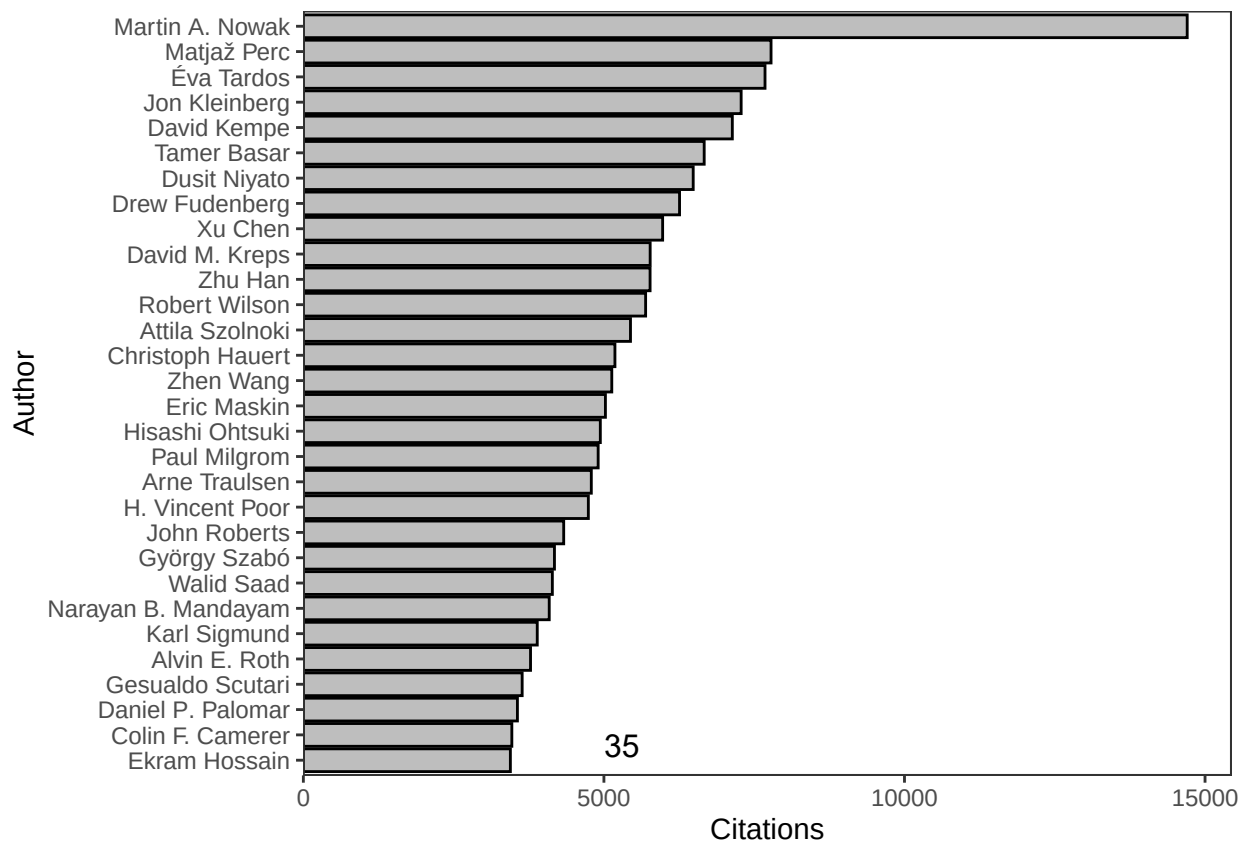
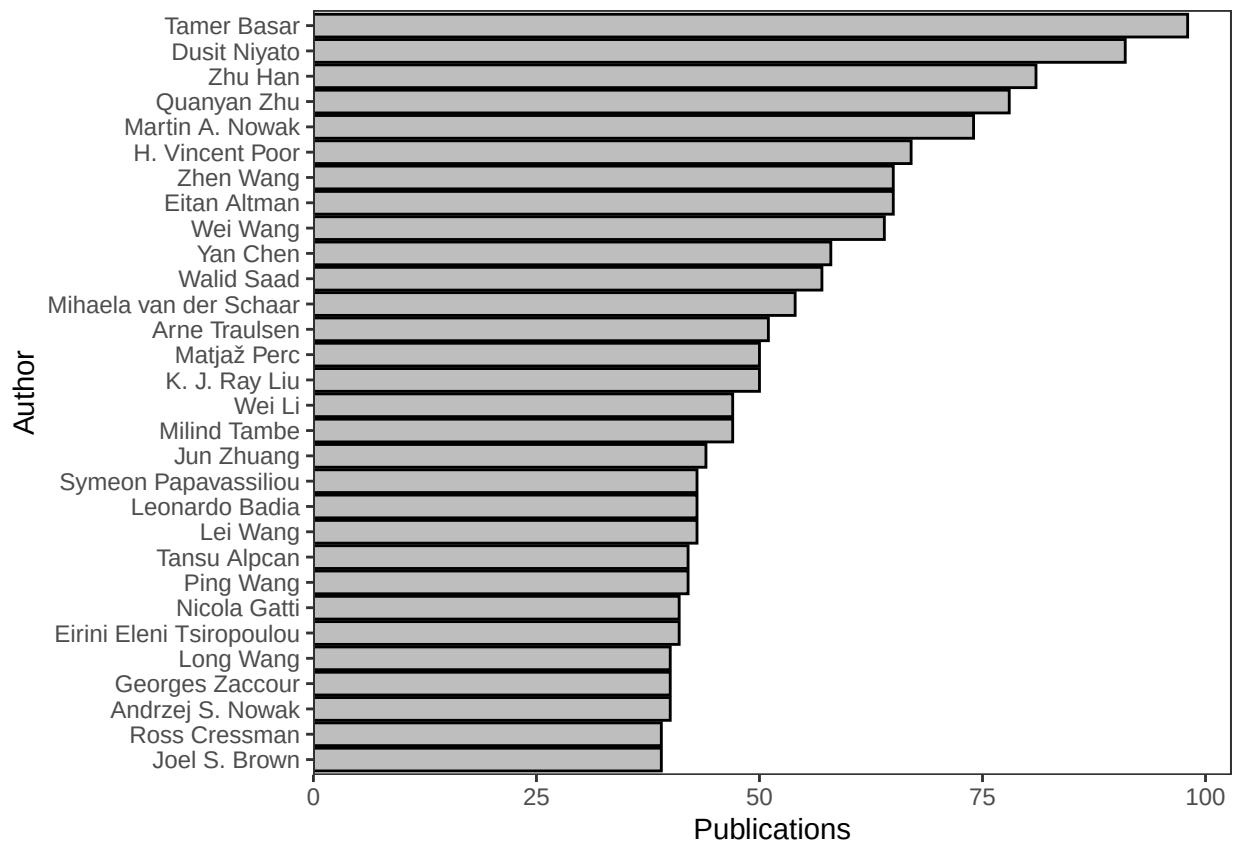


Subfields

- | | |
|--|--|
| — Aerospace Engineering | — Global and Planetary Change |
| — Artificial Intelligence | — Industrial and Manufacturing Engineering |
| — Automotive Engineering | — Information Systems |
| — Building and Construction | — Management Information Systems |
| — Civil and Structural Engineering | — Management Science and Operations Research |
| — Computational Theory and Mathematics | — Marketing |
| — Computer Networks and Communications | — Modeling and Simulation |
| — Control and Systems Engineering | — Ocean Engineering |
| — Ecology, Evolution, Behavior and Systematics | — Political Science and International Relations |
| — Economics and Econometrics | — Public Health, Environmental and Occupational Health |
| — Electrical and Electronic Engineering | — Safety Research |
| — Finance | — Sociology and Political Science |
| — General Decision Sciences | — Statistical and Nonlinear Physics |
| — General Economics, Econometrics and Finance | — Strategy and Management |
| — Genetics | — Transportation |

6. Authors Overview

6.1 Top Authors



6.2 Top Authors per Domain

Table 10: Top Authors in Terms of Citations per Domain

Health Sciences	Life Sciences	Physical Sciences	Social Sciences
Gerry Mackie (765)	Mart R. Gross (1681)	Jon Kleinberg (7136)	Martin A. Nowak (13532)
Alison P. Galvani (519)	Joel S. Brown (1455)	David Kempe (7132)	Matjaž Perc (7535)
Gretchen B. Chapman (519)	Hanna Kokko (1265)	Éva Tardos (7088)	Drew Fudenberg (6258)
Timothy C. Reluga (354)	Rufus A. Johnstone (1262)	Dusit Niyato (6381)	David M. Kreps (5769)
Don M. Hugie (260)	Karl Sigmund (1194)	Tamer Başar (5852)	Robert Wilson (5694)

Table 11: Top Authors in Terms of Publications per Domain

Health Sciences	Life Sciences	Physical Sciences	Social Sciences
Jan Rychtář (6)	Joel S. Brown (18)	Dusit Niyato (88)	Martin A. Nowak (70)
K. M. Ariful Kabir (5)	Ross Cressman (11)	Zhu Han (73)	Arne Traulsen (47)
Paolo Russu (5)	J. M. Cushing (9)	Tamer Başar (72)	Matjaž Perc (45)
Angelo Antoci (4)	Jan Rychtář (9)	Quanyan Zhu (66)	Zhen Wang (45)
Chris T. Bauch (4)	Gordon G. McNickle (7)	H. Vincent Poor (60)	Andrzej S. Nowak (39)

6.3 Top Authors per Field

Table 12: Top Authors in Terms of Citations per Field

Field	Authors		
Business, Management and Accounting	Arvind Parkhe (3117)	Rajiv Lal (1604)	Debabrata Ghosh (1411)
Computer Science	Dusit Niyato (4837)	Xu Chen (4786)	Narayan B. Mandayam (3411)
Decision Sciences	Robert Wilson (3160)	Eric Maskin (3152)	David M. Kreps (3101)
Economics, Econometrics and Finance	Charles F. Manski (2052)	Dilip Abreu (2047)	David Pearce (2005)
Engineering	Vincent W. S. Wong (3052)	Tamer Başar (2956)	Robert Schober (2932)
Mathematics	Kenneth J. Pienta (508)	David Axelrod (451)	Robert Axelrod (451)
Physics and Astronomy	Jon Kleinberg (7111)	David Kempe (7088)	Éva Tardos (7088)
Psychology	Herbert Gintis (411)	Geoffrey F. Miller (399)	Peter M. Todd (368)
Social Sciences	Martin A. Nowak (13206)	Matjaž Perc (7529)	Attila Szolnoki (5442)

Table 13: Top Authors in Terms of Publications per Field

Field	Authors		
Business, Management and Accounting	Salma Karray (19)	Ata Allah Taleizadeh (18)	Ruiliang Yan (18)
Computer Science	Dusit Niyato (49)	Tamer Başar (36)	Charles Kamhoua (32)
Decision Sciences	Eilon Solan (21)	Jason R. Marden (21)	János Flesch (21)
Economics, Econometrics and Finance	Andrzej S. Nowak (38)	René van den Brink (35)	S.H. Tijs (30)
Engineering	Walid Saad (42)	Zhu Han (41)	Dusit Niyato (38)
Mathematics	K. M. Ariful Kabir (7)	Andrzej Świerniak (5)	Michał Krześlak (5)
Physics and Astronomy	Yan Chen (15)	K. J. Ray Liu (9)	Azhar Iqbal (8)
Psychology	Kevin McGee (4)	Teong Leong Chuah (3)	Timothy Merritt (3)
Social Sciences	Martin A. Nowak (67)	Arne Traulsen (47)	Matjaž Perc (44)

6.4 Top Authors per Subfield

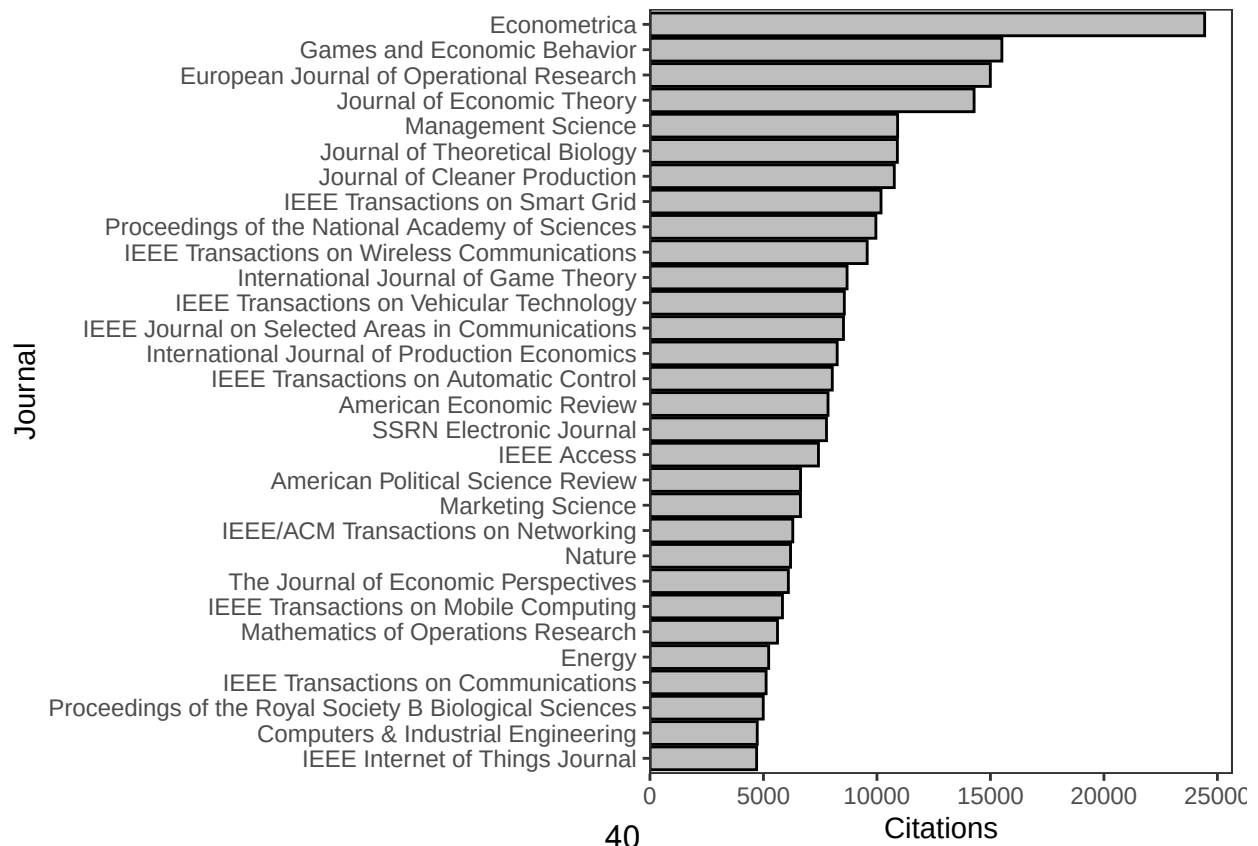
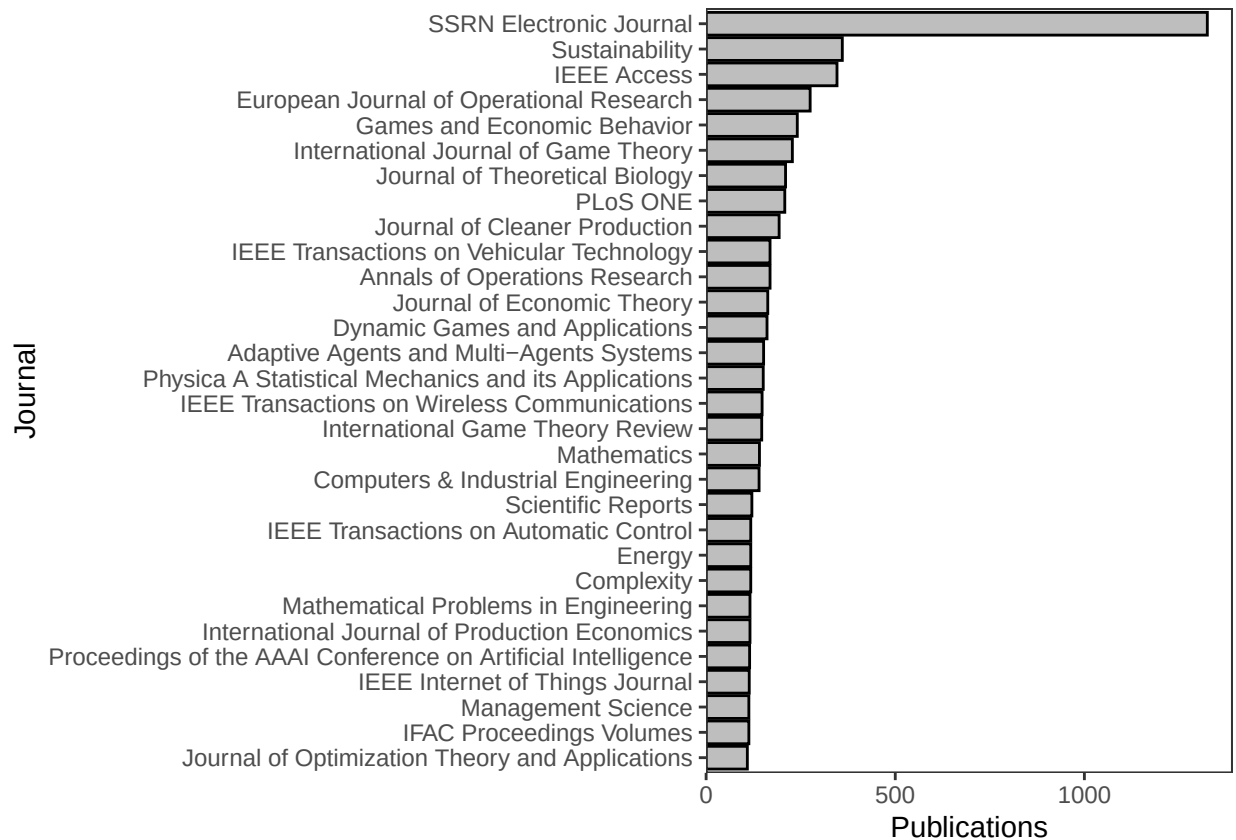
Table 14: Top Authors in Terms of Citations per Subfield

Subfield	Authors		
Aerospace Engineering	Arumugam Nallanathan (527)	Jingjing Cui (527)	Yuanwei Liu (527)
Artificial Intelligence	Xu Chen (2619)	Lei Jiao (2552)	Wenzhong Li (2552)
Automotive Engineering	Xiaoxiang Na (476)	David J. Cole (331)	Adam Millard□Ball (291)
Building and Construction	Zhi Yang (275)	Jian Zuo (234)	Guangdong Wu (214)
Civil and Structural Engineering	Milind Tambe (1178)	Jun Zhuang (943)	Vicki M. Bier (600)
Computational Theory and Mathematics	Frank L. Lewis (681)	Kyriakos G. Vamvoudakis (508)	Anne Condon (501)
Computer Networks and Communications	Narayan B. Mandayam (3349)	Dusit Niyato (2861)	D.J. Goodman (2571)
Control and Systems Engineering	Tamer Başar (1998)	P. Bernhard (1066)	Daniel E. Quevedo (928)
Ecology, Evolution, Behavior and Systematics	Mart R. Gross (1681)	Hanna Kokko (1265)	Rufus A. Johnstone (126)
Economics and Econometrics	Charles F. Manski (2052)	Dilip Abreu (2047)	David Pearce (2005)
Electrical and Electronic Engineering	Vincent W. S. Wong (3052)	Robert Schober (2932)	Daniel P. Palomar (2727)
Finance	Agatha Murgoci (1072)	Tomas Björk (1068)	Han Smit (646)
General Decision Sciences	Colin F. Camerer (1305)	Teck□Hua Ho (1002)	Dražen Prelec (303)
General Economics, Econometrics and Finance	Thomas J. Sargent (673)	William A. Bomberger (673)	Francisco J. Ruge-Murcia
Genetics	Karl Sigmund (1194)	Martin A. Nowak (1162)	Lin Chao (795)
Global and Planetary Change	Ray Dybzinski (426)	U. Rashid Sumaila (317)	Jacob Johansson (291)
Industrial and Manufacturing Engineering	Yong Wang (331)	Maozeng Xu (301)	Xiaolei Ma (301)
Information Systems	Dusit Niyato (1654)	Ping Wang (1527)	Wenbo Wang (1509)
Management Information Systems	Albert Y. Ha (901)	Greys Sošić (678)	Mahesh Nagarajan (678)
Management Science and Operations Research	Robert Wilson (3160)	Eric Maskin (3152)	David M. Kreps (3101)
Marketing	Rajiv Lal (1604)	C. Lanier Benkard (869)	Jonathan Levin (869)
Modeling and Simulation	Kenneth J. Pienta (508)	David Axelrod (451)	Robert Axelrod (451)
Ocean Engineering	Kaveh Madani (591)	Reza Kerachian (479)	Armaghan Abed□Elmdou
Political Science and International Relations	Alberto Alesina (823)	Wolfgang Mayer (666)	Robert Powell (653)
Public Health, Environmental and Occupational Health	Gerry Mackie (765)	Don M. Hugie (260)	Lawrence M. Dill (260)
Safety Research	Martin Dufwenberg (2977)	David M. Kreps (2512)	John Roberts (2512)
Sociology and Political Science	Martin A. Nowak (12722)	Matjaž Perc (7465)	Attila Szolnoki (5436)
Statistical and Nonlinear Physics	Jon Kleinberg (7111)	David Kempe (7088)	Éva Tardos (7088)
Strategy and Management	Arvind Parkhe (3117)	Debabrata Ghosh (1404)	Janat Shah (1392)
Transportation	Michael G.H. Bell (596)	Nicolás E. Stier□Moses (412)	Andreas S. Schulz (386)

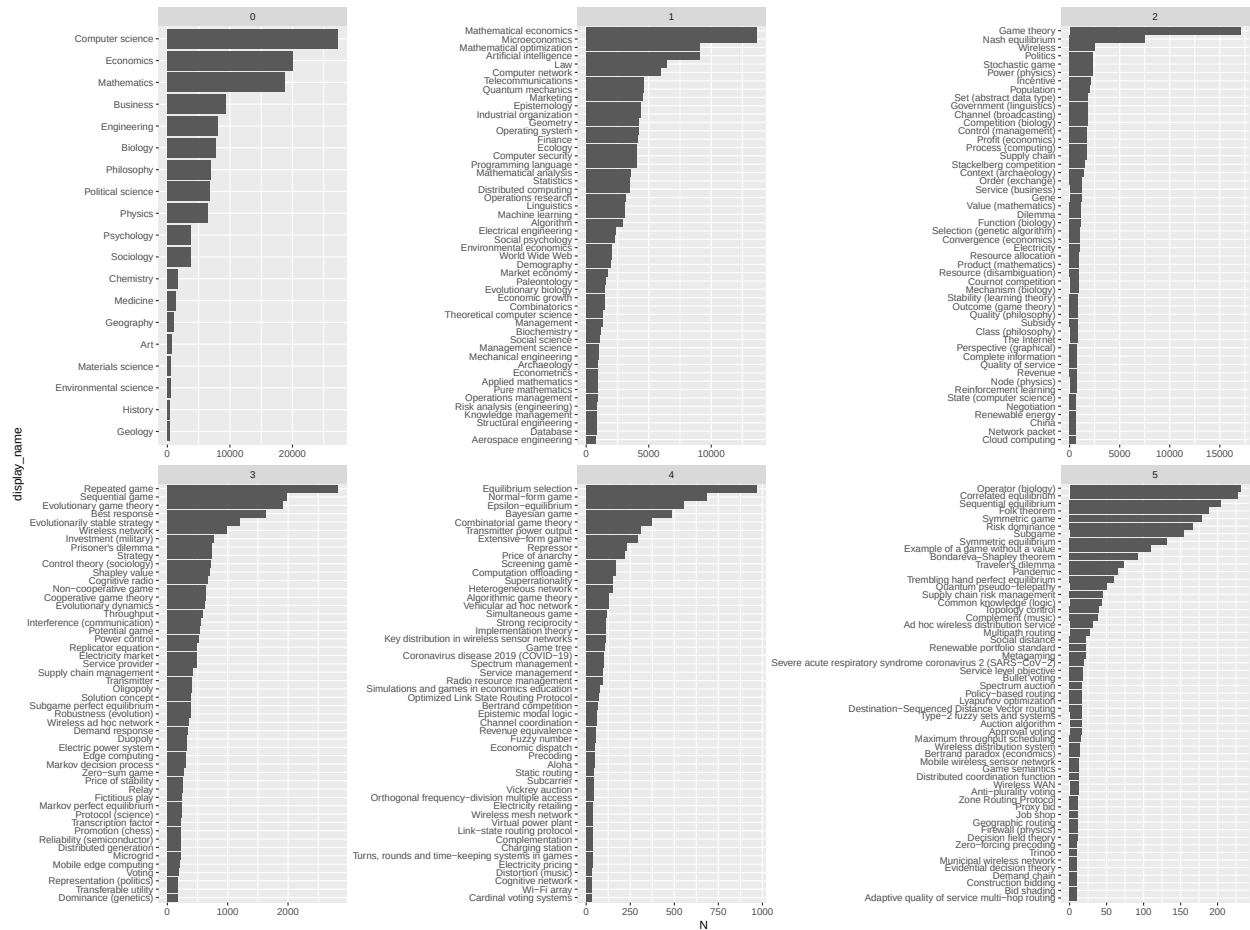
Table 15: Top Authors in Terms of Publications per Subfield

	Authors
Subfield	

7. Journal Overview



8. Distribution of Concepts



9. Keywords

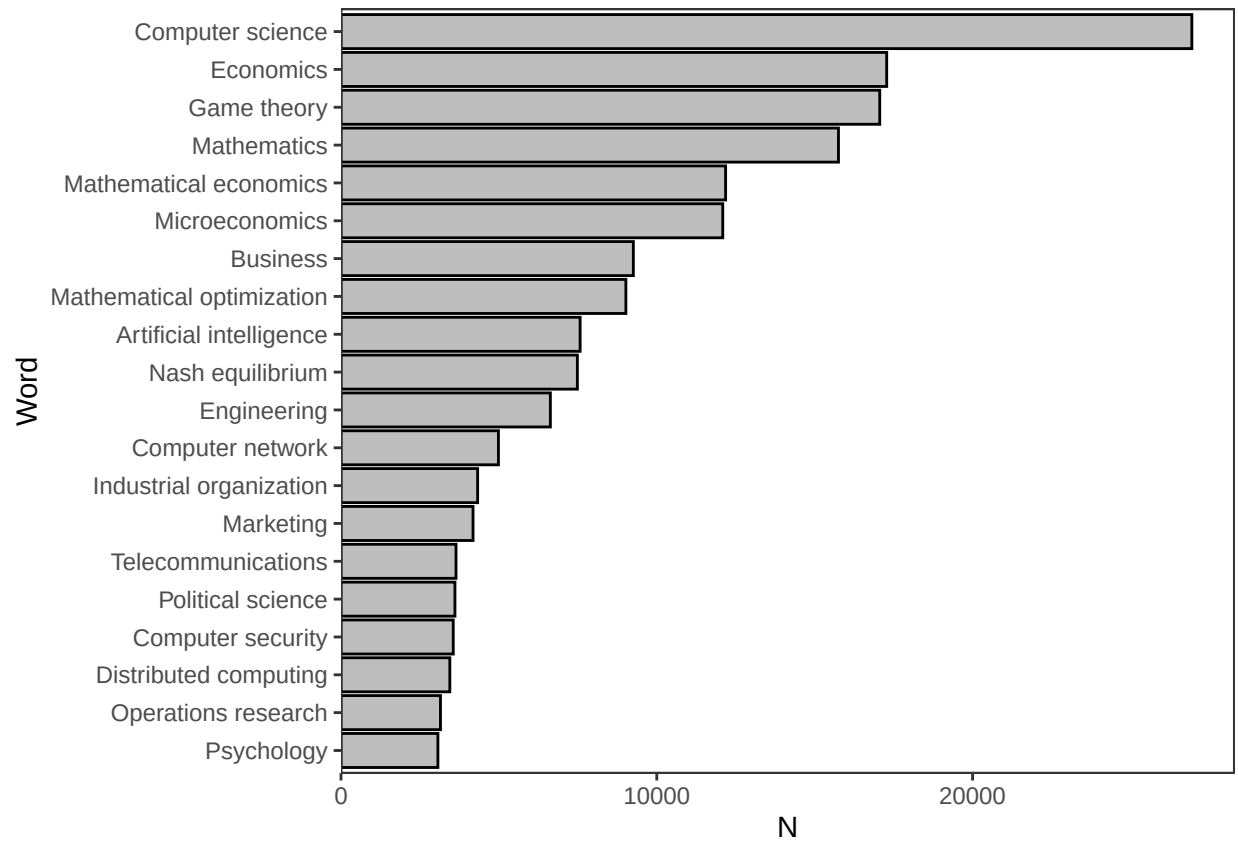


Table 16: Top Keywords

keywords	N
Computer science	26944
Economics	17281
Game theory	17061
Mathematics	15753
Mathematical economics	12180
Microeconomics	12089
Business	9258
Mathematical optimization	9021
Artificial intelligence	7572
Nash equilibrium	7485
Engineering	6630
Computer network	4985
Industrial organization	4327
Marketing	4181
Telecommunications	3642
Political science	3608
Computer security	3554
Distributed computing	3446
Operations research	3150
Psychology	3069